

A parity factorization and sharp stabilization for Cigler's Hankel polynomials

A proposed proof of Conjecture 16 in arXiv:2111.14492v3

Part II (added 29 September 2026): rectangular Schur structure
and sharp recurrences

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Abstract

We study the shifted Hankel determinants of Cigler's polynomial extension $c_n(t)$ of the middle binomial coefficients. An even-odd decomposition factors every normalized determinant into two principal minors of powers of simple tridiagonal matrices. This gives a proof of polynomiality, strict coefficient positivity, reciprocity, and the exact degree. An elementary alternant evaluation of the two factors proves the stable-coefficient assertions of Conjecture 16. More precisely, if $a = \lfloor k/2 \rfloor$, $b = \lfloor (k-1)/2 \rfloor$, and $\mathcal{H}_k(n; t) = (-1)^k \binom{n}{2} t^{-\binom{n}{2}} \det(c_{k+i+j}(t))_{0 \leq i, j < n}$, then for $k \geq 2$,

$$\mathcal{H}_k(n; t) = \frac{1}{(1-t)^a (1-t^2)^{ab}} - \binom{n+a}{a-1} t^{n+1} + O(t^{n+2}).$$

Thus the conjectured stable range is optimal, and its first defect has a closed formula. We also obtain fixed-size determinant evaluations, an explicit nonminimal recurrence, and convergence on compact subsets of the open unit disk. Exact integer and polynomial verification programs accompany the article. A stronger rectangular-Schur identity suggested by computation is separated from the proved results and is not used in any proof. [Added 29 September 2026, batch 56: Part II proves this identity and the exact minimal recurrence.]

Status and scope. This is an AI-assisted research draft with a full mathematical argument, not an independently refereed or proof-assistant-checked result. The target is Conjecture 16 in Section 6 of Cigler's paper, not Conjectures 13–15 in Section 5, which are already represented in the supplied manifest. A targeted literature search did not locate a later solution of this specific conjecture; that is not a guarantee of priority.

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1 The problem and the main result

1.1 Why this problem is not a duplicate

The supplied manifest [3] catalogues a report entitled *Product formulas for ballot-polynomial Hankel determinants*, concerning Conjectures 13–15 in Section 5 of Cigler's paper [1]. Those conjectures concern the moment family denoted there by $a_n(t)$. Here the family is $c_n(t)$ from equation (6), and the target is *Conjecture 16 in Section 6*, on pages 21–22 of version 3. The relationship is methodological—polynomial moment sequences and their Hankel determinants—but the input moments and the numbered problem differ. We do not assume that any proof catalogued in the manifest is correct.

The source continues to present this statement as a conjecture in the version located during the search. The arXiv record identifies version 3 as dated 30 December 2021 [1]. The bounded status audit in Section 11 explains what was and was not established about subsequent literature.

1.2 Definitions and conventions

A binomial coefficient $\binom{r}{j}$ is zero unless $0 \leq j \leq r$. Set $c_0(t) = 1$, and for $r \geq 1$ define

$$c_r(t) = \sum_{j=1}^r \binom{\lfloor (r-1)/2 \rfloor}{\lfloor (j-1)/2 \rfloor} \binom{\lfloor r/2 \rfloor}{\lfloor j/2 \rfloor} t^{j-1}. \quad (1.1)$$

This is Cigler's equation (6), with its zero $j = 0$ term omitted so that no negative power of t is displayed. For example,

$$\begin{array}{lll} c_0 = 1, & c_1 = 1, & c_2 = 1 + t, \\ c_3 = 1 + t + t^2, & c_4 = 1 + 2t + 2t^2 + t^3, & c_5 = 1 + 2t + 4t^2 + 2t^3 + t^4. \end{array}$$

For $k, n \geq 0$, write

$$D_k(n; t) = \det(c_{k+i+j}(t))_{0 \leq i, j < n}, \quad D_k(0; t) = 1. \quad (1.2)$$

For $k \geq 1$ initially define, in $\mathbb{Q}(t)$,

$$\mathcal{H}_k(n; t) = (-1)^{k \binom{n}{2}} \frac{D_k(n; t)}{t^{\binom{n}{2}}}. \quad (1.3)$$

Polynomiality, including a well-defined value at $t = 0$, will be proved. It is important not to substitute $t = 0$ into the quotient before this step.

Let

$$a = \left\lfloor \frac{k}{2} \right\rfloor, \quad b = \left\lfloor \frac{k-1}{2} \right\rfloor, \quad \mathcal{F}_k(t) = \frac{1}{(1-t)^a (1-t^2)^{ab}}. \quad (1.4)$$

An expression $O(t^r)$ in a formal-series identity means an element of $t^r \mathbb{Q}[[t]]$; it has no analytic meaning unless one is explicitly stated.

Theorem 1.1 (Conjecture 16, with a sharp first defect). *For every $k \geq 1$ and $n \geq 0$, $\mathcal{H}_k(n; t)$ is a polynomial in $\mathbb{Z}[t]$ of exact degree $(k-1)n$. Its constant and leading coefficients are both 1, every coefficient between them is strictly positive, and*

$$\mathcal{H}_k(n; t) = t^{(k-1)n} \mathcal{H}_k(n; t^{-1}). \quad (1.5)$$

For $k = 1$, $\mathcal{H}_1(n; t) = 1$. For $k \geq 2$,

$$\mathcal{H}_k(n; t) = \mathcal{F}_k(t) - \binom{n+a}{a-1} t^{n+1} + O(t^{n+2}). \quad (1.6)$$

Consequently,

$$[t^j]\mathcal{H}_k(n; t) = [t^j]\mathcal{F}_k(t) \quad (0 \leq j \leq n), \quad (1.7)$$

and $n + 1$ is the first degree of disagreement for every $k \geq 2$.

For $k = 2a$, equation (1.7) has denominator $(1 - t)^a(1 - t^2)^{a^2 - a}$; for $k = 2a + 1$ it has denominator $(1 - t)^a(1 - t^2)^{a^2}$. These are precisely the two stable-coefficient assertions in the source. The first-defect formula (1.6) is a refinement, not part of the original statement.

1.3 Proof architecture

The proof has three independent ingredients. First, the relation $c_{2r+2} = (1 + t)c_{2r+1}$ turns a Hankel matrix into a two-by-two block matrix whose determinant factors by row operations. Second, the resulting factors are principal minors of powers of coefficientwise totally nonnegative tridiagonal matrices. This establishes positivity and degree without any alternant or Schur identity. Finally, small alternant determinants give the first omitted terms of the two factors and hence the exact stable range. All determinant identities needed below are proved in the article.

[Added 29 September 2026, batch 56: Sections 1–11 and Appendices A and B are this report as of 20 September 2026 (Part I). Part II (Sections 12–22 and Appendices C and D; provenance in Section 12.1) is a later manuscript written against this text. It proves the experimental formula of Section 10, determines the exact minimal recurrence that replaces the bound of Proposition 9.1, and re-derives the parity factorization by a second route. It changes no statement of Part I.]

2 Two auxiliary moment sequences

Put $B = 1 + t^2$ and $s = 1 + t$, so that $s^2 = B + 2t$. For a Laurent polynomial in z , CT_z denotes its constant coefficient. Define

$$Q(z) = B + t(z + z^{-1}) = (1 + tz)(1 + t/z), \quad u_r = \text{CT}_z(1 + z)Q(z)^r, \quad v_r = \text{CT}_z(1 - z^2)Q(z)^r. \quad (2.1)$$

Both u_r and v_r belong to $\mathbb{Z}[t]$.

Lemma 2.1 (The pairing relation). *For every $r \geq 0$,*

$$c_{2r+1} = u_r, \quad c_{2r+2} = su_r, \quad s^2 u_r - u_{r+1} = tv_r. \quad (2.2)$$

Proof. Expanding $Q(z)^r = (1 + tz)^r(1 + t/z)^r$ gives

$$u_r = \sum_{j=0}^r \binom{r}{j}^2 t^{2j} + \sum_{j=0}^{r-1} \binom{r}{j} \binom{r}{j+1} t^{2j+1}. \quad (2.3)$$

Separating the even and odd indices of (1.1) gives $c_{2r+1} = u_r$. The coefficients of t^{2j} and t^{2j+1} in $(1 + t)u_r$ are respectively

$$\binom{r}{j} \left(\binom{r}{j} + \binom{r}{j-1} \right), \quad \binom{r}{j} \left(\binom{r}{j} + \binom{r}{j+1} \right).$$

Pascal's identity identifies these with the coefficients of c_{2r+2} . For the last assertion, observe that

$$(s^2 - Q(z))(1 + z) = t(1 + z - z^2 - z^{-1}).$$

Since $Q(z) = Q(z^{-1})$, the constant terms of $zQ(z)^r$ and $z^{-1}Q(z)^r$ agree. The unwanted two terms cancel, leaving tv_r . $\square$

2.1 Tridiagonal realizations

Let $e_0, e_1, \dots$ be the standard basis, let $Se_j = e_{j+1}$ be the unilateral shift, and let $E_{00} = e_0 e_0^\top$. For $\varepsilon = 0, 1$ put

$$M_\varepsilon = BI + t(S + S^\top) + \varepsilon t E_{00}. \quad (2.4)$$

Thus every off-diagonal entry adjacent to the diagonal is t , every diagonal entry is $1 + t^2$, and for M_1 the first diagonal entry has the additional summand t . These are infinite banded matrices: every entry of a fixed power and every minor used below is a finite calculation.

Lemma 2.2 (Moment realization). *For $r \geq 0$,*

$$u_r = (M_1^r)_{00}, \quad v_r = (M_0^r)_{00}. \quad (2.5)$$

Proof. For the first identity use Laurent polynomials $f_h = z^h + z^{-h-1}$, $h \geq 0$. Multiplication by $z + z^{-1}$ sends f_0 to $f_0 + f_1$ and sends f_h , $h > 0$, to $f_{h-1} + f_{h+1}$. Also $\text{CT } f_h$ is 1 for $h = 0$ and 0 otherwise. Multiplication by $Q(z)$ in this basis therefore has matrix M_1 , and $\text{CT } Q(z)^r f_0 = (M_1^r)_{00}$. Replacing z by z^{-1} gives the constant term defining u_r .

For the second identity use $g_h = z^h - z^{-h-2}$. Multiplication by $z + z^{-1}$ sends g_0 to g_1 and, for $h > 0$, sends g_h to $g_{h-1} + g_{h+1}$. Again $\text{CT } g_h = \delta_{h0}$. This gives M_0 and $\text{CT } Q(z)^r (1 - z^{-2}) = v_r$. $\square$

3 Hankel determinants as principal minors

For a matrix A , $A[m]$ means its principal submatrix on indices $0, \dots, m-1$, and $\det A[0] = 1$. Define

$$\mathcal{U}_a(m; t) = \det(M_1^a[m]), \quad \mathcal{V}_a(m; t) = \det(M_0^a[m]) \quad (a, m \geq 0). \quad (3.1)$$

Here $M_\varepsilon^a[m]$ means $(M_\varepsilon^a)[m]$, *not* $(M_\varepsilon[m])^a$. Paths may leave the first m indices and return before the a multiplications are complete.

Lemma 3.1 (The auxiliary Gram identity). *For $a, m \geq 0$,*

$$\det(u_{a+i+j})_{0 \leq i, j < m} = t^{m(m-1)} \mathcal{U}_a(m; t), \quad (3.2)$$

$$\det(v_{a+i+j})_{0 \leq i, j < m} = t^{m(m-1)} \mathcal{V}_a(m; t). \quad (3.3)$$

Proof. We prove both statements together. Write $M = M_\varepsilon$ and define a moment functional $L(X^r) = e_0^\top M^r e_0$. Let

$$p_0 = 1, \quad p_1 = X - B - \varepsilon t, \quad p_{j+1} = (X - B)p_j - t^2 p_{j-1} \quad (j \geq 1).$$

Induction gives $p_j(M)e_0 = t^j e_j$. Consequently,

$$L(p_i p_j) = t^{i+j} \delta_{ij}, \quad L(X^a p_i p_j) = t^{i+j} (M^a)_{ij}. \quad (3.4)$$

The change from $1, X, \dots, X^{m-1}$ to the monic polynomials $p_0, \dots, p_{m-1}$ is unit triangular. Changing both bases in the Hankel Gram matrix therefore preserves its determinant. Taking the factors t^i and t^j out of its rows and columns gives $t^{2 \sum_{i=0}^{m-1} i} = t^{m(m-1)}$, as required. This argument is an identity over $\mathbb{Z}[t]$ and does not require $t \neq 0$. $\square$

The normalized auxiliary quantities are thus polynomials before we use any rational evaluation of them. In particular $\mathcal{U}_0(m) = \mathcal{V}_0(m) = 1$ and $\mathcal{U}_a(0) = \mathcal{V}_a(0) = 1$.

4 The parity factorization

Theorem 4.1 (Four parity cases). *For $a, m \geq 0$,*

$$\mathcal{H}_{2a+1}(2m; t) = \mathcal{U}_a(m; t) \mathcal{V}_a(m; t), \quad (4.1)$$

$$\mathcal{H}_{2a+1}(2m+1; t) = \mathcal{U}_a(m+1; t) \mathcal{V}_a(m; t). \quad (4.2)$$

For $a \geq 1$ and $m \geq 0$,

$$\mathcal{H}_{2a}(2m; t) = \mathcal{U}_a(m; t) \mathcal{V}_{a-1}(m; t), \quad (4.3)$$

$$\mathcal{H}_{2a}(2m+1; t) = (1+t) \mathcal{U}_{a-1}(m+1; t) \mathcal{V}_a(m; t). \quad (4.4)$$

In particular the normalization in (1.3) is polynomial.

Proof. Reorder both rows and columns of the Hankel matrix by putting the even indices first. Applying the same permutation on both sides contributes no sign. Write $U_a(r)$ for the unnormalized matrix $(u_{a+i+j})_{0 \leq i, j < r}$ and $V_a(r)$ for the analogous matrix of v 's.

For an odd shift $2a+1$, the four blocks, with the appropriate rectangular sizes, are

$$\begin{pmatrix} (u_{a+i+j}) & (su_{a+i+j}) \\ (su_{a+i+j}) & (u_{a+i+j+1}) \end{pmatrix}.$$

Subtract s times the corresponding even row from each odd row. The lower left block becomes zero and the lower right block becomes $(-tv_{a+i+j})$, by Lemma 2.1. The upper left block has size m when $n = 2m$ and size $m+1$ when $n = 2m+1$. Therefore

$$D_{2a+1}(2m) = (-t)^m \det U_a(m) \det V_a(m),$$

$$D_{2a+1}(2m+1) = (-t)^m \det U_a(m+1) \det V_a(m).$$

Using Lemma 3.1, the total powers of t are respectively $2m^2 - m = \binom{2m}{2}$ and $2m^2 + m = \binom{2m+1}{2}$. Also $(-1)^m = (-1)^{\binom{n}{2}}$ for both $n = 2m$ and $n = 2m+1$. This proves the two odd-shift formulas.

For shift $2a$ and size $2m$, the reordered matrix is

$$\begin{pmatrix} sU_{a-1}(m) & U_a(m) \\ U_a(m) & sU_a(m) \end{pmatrix}.$$

Work temporarily in $\mathbb{Q}(t)$ and subtract s^{-1} times each bottom row from the matching top row. The upper right block becomes zero, and the upper left block is $(t/s)V_{a-1}(m)$. Hence

$$D_{2a}(2m) = t^m \det V_{a-1}(m) \det U_a(m).$$

For size $2m+1$, the even block has $m+1$ rows. Subtract s^{-1} times even row $i+1$ from odd row i , for $0 \leq i < m$. The lower left block vanishes; the lower right block becomes $(t/s)V_a(m)$, while the upper left block is $sU_{a-1}(m+1)$. Thus

$$D_{2a}(2m+1) = st^m \det U_{a-1}(m+1) \det V_a(m).$$

The Gram identities give exactly the same two normalization exponents as before. These are polynomial identities after denominators are cleared, so the temporary use of $1/s$ causes no exception at $t = -1$. $\square$

5 Strict positivity, degree, and reciprocity

For polynomials P, Q , write $P \succeq Q$ if $P - Q$ has nonnegative integer coefficients. A matrix is *coefficientwise totally nonnegative* if every finite minor, with row and column indices in increasing order, is a polynomial with nonnegative coefficients.

Lemma 5.1 (Total nonnegativity of the auxiliary matrices). *Both M_0 and M_1 , and all of their nonnegative integer powers, are coefficientwise totally nonnegative.*

Proof. Since $S^\top S = I$ for the unilateral shift,

$$M_0 = (I + tS^\top)(I + tS). \quad (5.1)$$

The two bidiagonal factors are totally nonnegative: a nonzero minor has an order-preserving matching of its selected rows and columns and has a nonnegative monomial weight. This can also be checked by expanding at the first selected row; the bidiagonal zero pattern prevents a negatively signed nonzero matching.

The Cauchy–Binet formula proves total nonnegativity of their product. Although the matrices are infinite, each of these sums is finite because the factors are banded. Next $M_1 = M_0 + tE_{00}$. A minor not selecting both row 0 and column 0 is unchanged. A minor selecting both acquires t times the minor obtained by deleting its first row and first column. The cofactor sign is positive because the selected indices are increasing. Thus adding this corner entry preserves coefficientwise total nonnegativity. Repeated Cauchy–Binet proves the assertion for powers. $\square$

Proposition 5.2 (Support and reciprocity of the factors). *For $a, m \geq 0$, $\mathcal{U}_a(m)$ and $\mathcal{V}_a(m)$ are reciprocal polynomials of exact degree $2am$, with constant and leading coefficients 1. For $a, m \geq 1$, every coefficient of $\mathcal{U}_a(m)$ is positive, whereas $\mathcal{V}_a(m)$ has a positive coefficient at every even degree between 0 and $2am$ and no odd-degree terms. More quantitatively,*

$$\mathcal{U}_a(m; t) \succeq (1 + t + \cdots + t^{2m})^a, \quad (5.2)$$

$$\mathcal{V}_a(m; t) \succeq (1 + t^2 + \cdots + t^{2m})^a. \quad (5.3)$$

Proof. The matrices satisfy $M_\varepsilon(t^{-1}) = t^{-2}M_\varepsilon(t)$. Taking the a th power and an m -row principal minor gives

$$\mathcal{U}_a(m; t) = t^{2am}\mathcal{U}_a(m; t^{-1}), \quad \mathcal{V}_a(m; t) = t^{2am}\mathcal{V}_a(m; t^{-1}).$$

At $t = 0$, the matrices are I , so both determinants have constant term 1. Their exact degrees and leading coefficients follow.

The tridiagonal determinant recurrence, with $B = 1 + t^2$, gives

$$\det M_1[m] = 1 + t + \cdots + t^{2m}, \quad \det M_0[m] = 1 + t^2 + \cdots + t^{2m}.$$

For example the second sequence has initial values $1, B$ and recurrence $P_m = BP_{m-1} - t^2P_{m-2}$; the first has initial values $1, B + t$ and the same recurrence. The displayed finite sums satisfy these recurrences.

Expand the principal minor of M_ε^a by repeated Cauchy–Binet. All terms are coefficientwise nonnegative. Selecting the same index set $\{0, \dots, m-1\}$ at every intermediate stage supplies the term $(\det M_\varepsilon[m])^a$, proving the two lower bounds. Finally $M_0(-t) = JM_0(t)J$, where $J = \text{diag}(1, -1, 1, -1, \dots)$. Its principal minors of powers are unchanged, so $\mathcal{V}_a(m)$ is even. $\square$

Proof of the positivity and reciprocity part of Theorem 1.1. Apply Theorem 4.1. The degrees in its four lines are

$$4am, \quad 2a(2m+1), \quad 2m(2a-1), \quad (2a-1)(2m+1),$$

which equal $(k-1)n$ in each case. All factors are reciprocal, including $1+t$, and their constant terms are 1.

For strict positivity, the full consecutive support of a nonconstant $\mathcal{U}$ factor overlaps the translates arising from the even support of a $\mathcal{V}$ factor. More explicitly, an interval $\{0, \dots, L\}$ with $L \geq 1$, added to $\{0, 2, \dots, 2R\}$, covers every integer from 0 to $L+2R$. The only exceptional-looking even-shift case is $a=1$, n odd: there the factor $1+t$ supplies precisely the needed interval of length one. The cases $n=0$ and $k=1$ give the constant polynomial 1 directly. $\square$

6 A small alternant evaluation

Positivity did not require symmetric-function identities. The stabilization proof will also be elementary: the relevant formulas are quotients of explicit alternating determinants. This section supplies their derivation, including the classical moment determinant identity that is often used for this purpose; compare the general framework in [2].

6.1 The Christoffel determinant identity

Lemma 6.1 (Moment modification). *Let L be a moment functional over a characteristic-zero field, with monic orthogonal polynomials p_j of nonzero norms $h_j = L(p_j^2)$. Put $H_m = \det(L(X^{i+j}))_{0 \leq i,j < m}$. For independent parameters $\eta_1, \dots, \eta_a$,*

$$\frac{\det(L(X^{i+j} \prod_{r=1}^a (X - \eta_r)))_{0 \leq i,j < m}}{H_m} = (-1)^{am} \frac{\det(p_{m+j-1}(\eta_i))_{1 \leq i,j \leq a}}{\Delta(\eta)}, \quad (6.1)$$

where $\Delta(\eta) = \prod_{i < j} (\eta_j - \eta_i)$.

Proof. Form an $(m+a)$ -square matrix whose first m rows, indexed by r , are $L(X^r X^j)$ for $0 \leq j < m+a$, and whose last a rows are η_i^j . Change the column polynomials from the monomials to $p_0, \dots, p_{m+a-1}$. The change is unit triangular. Orthogonality makes the upper right block zero, and the determinant of the upper left block is $\prod_{r=0}^{m-1} h_r = H_m$. Thus this determinant is $H_m \det(p_{m+j-1}(\eta_i))$.

Alternatively change the column polynomials to $1, X, \dots, X^{a-1}, P, XP, \dots, X^{m-1}P$, where $P = \prod_r (X - \eta_r)$. This is again a monic unit-triangular change. Move the last a rows to the top, contributing $(-1)^{am}$. The top left block is a Vandermonde matrix and the top right block is zero. The bottom right block is the modified moment matrix on the left of (6.1). Equating the two evaluations proves the identity. $\square$

6.2 The two explicit numerators

For $a, m \geq 0$ and $\delta \in \{0, 1\}$, define

$$N_{a,m}^\delta(t) = \det \left(\binom{j-1}{i-1} t^{j-i} - \binom{2m+2a-j+\delta}{i-1} t^{2m+2a-j+\delta-i+1} \right)_{1 \leq i,j \leq a}. \quad (6.2)$$

The first summand is interpreted as zero when $j < i$, without evaluating its negative power. An empty determinant is 1.

Theorem 6.2 (Fixed-size evaluations). *For all $a, m \geq 0$,*

$$\mathcal{U}_a(m; t) = \frac{N_{a,m}^0(t)}{(1-t)^a(1-t^2)^{\binom{a}{2}}}, \quad (6.3)$$

$$\mathcal{V}_a(m; t) = \frac{N_{a,m}^1(t)}{(1-t^2)^{\binom{a+1}{2}}}. \quad (6.4)$$

These identities hold in $\mathbb{Q}(t)$ and determine the polynomials already constructed in (3.1); apparent singularities at $t = 1$ or $t = -1$ are removable. The determinant orders are a , independently of m .

Proof. Work first over $\mathbb{Q}(t)$. Let $C_0(y) = 1$, $C_1(y) = y$, and $C_{j+1}(y) = yC_j(y) - C_{j-1}(y)$; also put $C_{-1} = 0$. The orthogonal polynomials in Lemma 3.1 are

$$p_j^\varepsilon(X) = t^j \left[C_j \left(\frac{X-B}{t} \right) - \varepsilon C_{j-1} \left(\frac{X-B}{t} \right) \right].$$

Define monic polynomials $A_j^\varepsilon(y) = C_j(y) + \varepsilon C_{j-1}(y)$ and put

$$\eta_i = B - t(x_i + x_i^{-1}).$$

Then $p_j^\varepsilon(\eta_i) = (-t)^j A_j^\varepsilon(x_i + x_i^{-1})$. The elementary recurrence gives

$$A_j^0(x + x^{-1}) = \frac{x^{j+1} - x^{-j-1}}{x - x^{-1}}, \quad (6.5)$$

$$A_j^1(x + x^{-1}) = \frac{x^{j+1/2} - x^{-j-1/2}}{x^{1/2} - x^{-1/2}} = \sum_{r=-j}^j x^r. \quad (6.6)$$

The half powers in the middle expression cancel; the expression on the right is a Laurent polynomial.

Since the A_j^ε are monic in y , their first a evaluation columns have determinant $\Delta(y)$, where $y_i = x_i + x_i^{-1}$. Apply Lemma 6.1, cancel the column powers of $-t$ and the Vandermonde scale, and then specialize every η_i to zero. Choosing every $x_i = t$ gives

$$\det(M_\varepsilon^a[m]) = t^{am} \frac{\det(A_{m+j-1}^\varepsilon(x_i + x_i^{-1}))}{\det(A_{j-1}^\varepsilon(x_i + x_i^{-1}))} \Big|_{x_1=\dots=x_a=t}. \quad (6.7)$$

The equal-parameter value is the limit of an alternating quotient, not a quotient of two separately substituted zero determinants.

Reverse all columns in the numerator and denominator of this quotient and use (6.5)–(6.6). For $\delta = 1 - \varepsilon$, multiplication of row i by the appropriate power of x_i gives

$$\left(\prod_{i=1}^a x_i \right)^m \frac{\det(A_{m+j-1}^\varepsilon(x_i + x_i^{-1}))}{\det(A_{j-1}^\varepsilon(x_i + x_i^{-1}))} = \frac{\det(x_i^{j-1} - x_i^{2m+2a-j+\delta})}{\det(x_i^{j-1} - x_i^{2a-j+\delta})}. \quad (6.8)$$

At $x_i = t$, the prefactor on the left is exactly t^{am} .

The denominator on the right has the factorization

$$\det(x_i^{j-1} - x_i^{2a-j+\delta}) = \Delta(x) \prod_{i=1}^a (1 - x_i^{1+\delta}) \prod_{i < j} (1 - x_i x_j). \quad (6.9)$$

Here is a direct verification. Alternation supplies $\Delta(x)$. At $x_i = 1$ a row vanishes, and for $\delta = 1$ it also vanishes at $x_i = -1$. At $x_j = x_i^{-1}$ the two rows are proportional, with proportionality

factor $-x_i^{-(2a-1+\delta)}$, so every $1 - x_i x_j$ divides the determinant. These factors have exactly its total degree. Its lowest homogeneous term is $\Delta(x)$, which fixes the multiplicative constant as 1.

Finally, for polynomials f_j , Taylor expansion gives the confluent identity

$$\frac{\det(f_j(x_i))}{\Delta(x)} \Big|_{x_1=\dots=x_a=t} = \det \left(\frac{f_j^{(i-1)}(t)}{(i-1)!} \right)_{1 \leq i, j \leq a}. \quad (6.10)$$

Indeed the lowest alternating term in the Taylor expansion around t uses the distinct exponents $0, 1, \dots, a-1$. Apply (6.10) to the numerator in (6.8). It gives exactly (6.2). The remaining factors in (6.9), after equal specialization, give the denominators in (6.3) and (6.4). $\square$

Remark 6.3. The quotients before equal specialization are familiar rectangular orthogonal and symplectic character alternants. No character theory, branching formula, or bounded Schur-sum identity is being imported: the moment identity, the recurrence, the denominator factorization, and the confluent limit above give a complete determinant proof.

7 Sharp stable coefficients

Lemma 7.1 (First nonconstant numerator term). *For $a \geq 1$, $m \geq 0$, and $\delta \in \{0, 1\}$, put $r = 2m + 1 + \delta$. Then*

$$N_{a,m}^\delta(t) = 1 - \binom{a+r-1}{a-1} t^r + O(t^{r+1}). \quad (7.1)$$

Proof. Before equal specialization, expand $\det(x_i^{j-1} - x_i^{2m+2a-j+\delta})$ by choosing either the low or the high monomial in each column. If the high monomial is selected in column j , the total degree increases, relative to $\Delta(x)$, by

$$2m + 2a - 2j + 1 + \delta = r + 2(a - j).$$

The all-low choice is $\Delta(x)$. The unique choice of smallest positive additional degree is to select only the high monomial in column a ; its sign is negative and its additional degree is r . Every other choice has strictly larger additional degree.

After division by $\Delta(x)$ and equal specialization, the coefficient of this first correction can be computed directly from (6.10). The column exponents are $0, 1, \dots, a-2, a-1+r$. The first $a-1$ columns are triangular with unit diagonal, so the remaining determinant is $\binom{a-1+r}{a-1} t^r$. This proves the assertion, including $a = 1$. $\square$

Define the formal series

$$U_a^\infty(t) = \frac{1}{(1-t)^a (1-t^2)^{\binom{a}{2}}}, \quad V_a^\infty(t) = \frac{1}{(1-t^2)^{\binom{a+1}{2}}}. \quad (7.2)$$

Their constant terms are 1. For $a = 0$ both are exactly 1. Theorem 6.2 and Lemma 7.1 imply, for $a \geq 1$,

$$\mathcal{U}_a(m; t) = U_a^\infty(t) - \binom{2m+a}{a-1} t^{2m+1} + O(t^{2m+2}), \quad (7.3)$$

$$\mathcal{V}_a(m; t) = V_a^\infty(t) - \binom{2m+a+1}{a-1} t^{2m+2} + O(t^{2m+3}). \quad (7.4)$$

Multiplying by a series with constant term 1 does not change the first nonzero correction, which is why the same binomial coefficient appears in these formulas.

Completion of Theorem 1.1. For $n = 2m$ the earliest correction in both (4.1) and (4.3) comes from $\mathcal{U}_a(m)$, at degree $2m + 1 = n + 1$. Its magnitude is $\binom{2m+a}{a-1} = \binom{n+a}{a-1}$. The products of the corresponding limiting series are respectively

$$U_a^\infty V_a^\infty = \frac{1}{(1-t)^a(1-t^2)^{a^2}}, \quad U_a^\infty V_{a-1}^\infty = \frac{1}{(1-t)^a(1-t^2)^{a(a-1)}}.$$

For $n = 2m + 1$ the earliest correction in both (4.2) and (4.4) comes from $\mathcal{V}_a(m)$, at degree $2m + 2 = n + 1$, with magnitude $\binom{2m+a+1}{a-1} = \binom{n+a}{a-1}$. For the last parity case the limiting product simplifies as

$$(1+t)U_{a-1}^\infty V_a^\infty = \frac{1+t}{(1-t)^{a-1}(1-t^2)^{a^2-a+1}} = \frac{1}{(1-t)^a(1-t^2)^{a(a-1)}}.$$

All multiplying factors have constant term 1, so they do not alter the first correction coefficient. When $a = 1$, any factor with index $a - 1 = 0$ is identically 1, and the same argument applies. The positive binomial coefficient proves optimality of the range. The case $k = 1$ was already obtained directly from the factorization. $\square$

Corollary 7.2 (Coefficient extraction without a Hankel determinant). *For $k \geq 2$, $0 \leq j \leq n$, and a, b as in (1.4),*

$$[t^j]\mathcal{H}_k(n; t) = \sum_{\ell=0}^{\lfloor j/2 \rfloor} W(ab, \ell) W(a, j-2\ell), \quad (7.5)$$

where $W(q, r) = \binom{q+r-1}{r}$ for $q \geq 1$, and $W(0, r) = 1$ for $r = 0$ and 0 otherwise.

Proof. Expand $(1-t)^{-a}$ and $(1-t^2)^{-ab}$ and use (1.7). The convention for $W(0, r)$ handles $k = 2$ without an artificial negative upper binomial index. $\square$

8 Examples and checks on the statement

8.1 The first three shifts

The factorization immediately gives

$$\mathcal{H}_1(n; t) = 1, \quad \mathcal{H}_2(n; t) = 1 + t + \cdots + t^n. \quad (8.1)$$

For $k = 3$, insert $\mathcal{U}_1(m) = 1 + t + \cdots + t^{2m}$ and $\mathcal{V}_1(m) = 1 + t^2 + \cdots + t^{2m}$ in the two parity cases. Both simplify to

$$\mathcal{H}_3(n; t) = \frac{(1-t^{n+1})(1-t^{n+2})}{(1-t)(1-t^2)}. \quad (8.2)$$

These small-shift formulas are already printed in Cigler's paper; they serve as indexing and sign checks, not as new results.

8.2 A first-defect example

For $k = 4$, the limiting series is

$$\mathcal{F}_4(t) = \frac{1}{(1-t)^2(1-t^2)^2} = 1 + 2t + 5t^2 + 8t^3 + 14t^4 + \cdots.$$

At $n = 2$ the determinant gives

$$\mathcal{H}_4(2; t) = 1 + 2t + 5t^2 + 4t^3 + 5t^4 + 2t^5 + t^6.$$

The first three coefficients agree. At degree $3 = n + 1$ the deficit is $8 - 4 = 4 = \binom{2+2}{2-1}$, exactly as predicted. At $n = 3$,

$$\mathcal{H}_4(3; t) = 1 + 2t + 5t^2 + 8t^3 + 9t^4 + 9t^5 + 8t^6 + 5t^7 + 2t^8 + t^9,$$

and the first deficit is $14 - 9 = 5 = \binom{3+2}{2-1}$.

These examples also delimit the result. The coefficients 5, 4, 5 in $\mathcal{H}_4(2; t)$ have a strict valley. Thus positivity and reciprocity do *not* imply unimodality here. No unimodality or log-concavity assertion is included in Theorem 1.1.

8.3 Signs and exceptional parameters

For real $t > 0$, strict coefficient positivity gives the exact sign

$$\operatorname{sgn} D_k(n; t) = (-1)^{k \binom{n}{2}}. \quad (8.3)$$

At $t = 0$ the normalized polynomial has value 1, although the original Hankel determinant vanishes when $n \geq 2$. At $t = 1$ the source moments are the middle binomial coefficients. At $t = -1$ the factor $1 + t$ in (4.4) shows that every positive even-shift, odd-size determinant vanishes. These statements follow from polynomial identities; no argument uses an undefined specialization of a rational formula.

9 Algorithms and further consequences

9.1 Exact computation at fixed shift

Theorem 6.2 evaluates the factors through determinants of order at most $\lfloor k/2 \rfloor$, even when n is very large. A practical algorithm is to determine the parity of k, n , compute the one or two numerators in (6.2), divide by their proved exact denominators, and multiply according to Theorem 4.1. For $t = \pm 1$, use the principal-minor definition or take the polynomial limit instead of evaluating a zero denominator.

The reduction is in determinant *order*, not in bit complexity. The entries contain powers whose exponents grow with n , and an expanded answer has degree $(k - 1)n$. Once entries are constructed, ordinary dense elimination uses a cubic number of field operations in the small determinant order; large integers and polynomial output still carry their usual costs. For only the first $j \leq n$ coefficients, (7.5) avoids both determinant constructions.

9.2 A concrete, nonminimal recurrence

Proposition 9.1 (Rationality at each fixed shift). *For $k \geq 2$, set*

$$d_k = \begin{cases} (a - 1)^2, & k = 2a, \\ a(a - 1), & k = 2a + 1. \end{cases}$$

The series $\sum_{n \geq 0} \mathcal{H}_k(n; t)x^n$ is rational over $\mathbb{Q}(t)$, with a denominator dividing

$$\prod_{q=0}^{k-1} (1 - t^{2q}x^2)^{d_{k+1}}. \quad (9.1)$$

Equivalently, with E denoting forward shift in n ,

$$\prod_{q=0}^{k-1} (E^2 - t^{2q})^{d_{k+1}} \mathcal{H}_k(n; t) = 0. \quad (9.2)$$

The bound is not asserted to be minimal and is not a resolution of all the more precise denominator and numerator assertions in Conjecture 18.

Proof. Expand (6.2) column by column. A term selecting q high columns has the form $P(m; t)t^{2qm}$, where P is a polynomial in m with coefficients in $\mathbb{Q}(t)$. Its degree in m is at most $\sum_{i=1}^a (i-1) = \binom{a}{2}$, because the binomial in row i has degree at most $i-1$. The denominators in Theorem 6.2 do not depend on m . Thus each auxiliary factor is a finite sum of $m^j(t^{2q})^m$, with $0 \leq q \leq a$ and $j \leq \binom{a}{2}$.

The products in the two parity subsequences have $0 \leq q \leq k-1$. Their degree bounds are $2\binom{a}{2} = a(a-1)$ in the odd-shift case and $\binom{a}{2} + \binom{a-1}{2} = (a-1)^2$ in the even-shift case. Replacing m by $m+1$ changes neither bound. A polynomial of degree at most d times λ^m is annihilated by $(E_m - \lambda)^{d+1}$, as follows by taking successive finite differences. Interleaving the even and odd subsequences replaces E_m by E^2 , proving the claim. $\square$

[Added 29 September 2026, batch 56: in the notation of this proposition, the exact reduced denominator is $\prod_{q=0}^{k-1} (1 - t^q x)^{1 + \lfloor q(k-1-q)/2 \rfloor}$, with no cancellation over $\mathbb{Q}(t)$ or at any real $t \notin \{0, \pm 1\}$ (Theorem 12.2; Part II writes z for the variable x). Part II also gives the numerator degree and reciprocity sign (Theorem 18.2), the signed denominator for odd k (Theorem 18.3), and the reduced denominators at $t = 0, \pm 1$ (Theorem 19.2). Cigler's denominators (84) and (86) are minimal; the positive sign printed in his (85) is false when a is even, the correct sign being $(-1)^{a+1}$.]

For the unadjusted normalized determinant $d_k(n; t) = t^{-\binom{n}{2}} D_k(n; t)$, the even- k recurrence is unchanged. For odd k , the period-four sign replaces $1 - t^{2q}x^2$ in the displayed coarse denominator by $1 + t^{2q}x^2$.

9.3 Actual convergence, not just formal stabilization

Proposition 9.2 (Uniform convergence inside the unit disk). *Fix $k \geq 2$ and $0 < r < 1$. There is a constant $C_{k,r}$ such that, for all $n \geq 0$,*

$$\sup_{|t| \leq r} |\mathcal{H}_k(n; t) - \mathcal{F}_k(t)| \leq C_{k,r} (n+1) \binom{a}{2} r^{n+1}. \quad (9.3)$$

In particular the polynomials converge to $\mathcal{F}_k$ locally uniformly on $|t| < 1$.

Proof. In the expansion of $N_{a,m}^\delta - 1$, every nonzero term has degree at least $2m + 1 + \delta$ in t . There are finitely many determinant terms for fixed a , and each coefficient is bounded by a constant times $(m+1)\binom{a}{2}$, by the row-degree estimate just used. It follows, uniformly on $|t| \leq r$, that

$$N_{a,m}^\delta(t) - 1 = O_{a,r}((m+1)\binom{a}{2}r^{2m+1+\delta}).$$

The two fixed denominators in Theorem 6.2 are bounded away from zero on this disk. Hence the factors converge uniformly to (7.2), with these bounds, and in particular remain uniformly bounded as m varies. Subtract the limiting products in the four parity formulas. The earliest error has exponent $n+1$, exactly as in Section 7; the largest polynomial exponent is $\binom{a}{2}$. The finitely many small cases can be absorbed in $C_{k,r}$. $\square$

[Added 29 September 2026, batch 56: for fixed $0 < t < 1$ the exact rate is $\mathcal{F}_k(t) - \mathcal{H}_k(n; t) \sim n^{a-1}t^{n+1}/((a-1)!(1-t)^{b+1}(1-t^2)^{(a-1)b})$, with relative error $O(1/n)$ locally uniformly on $(0, 1)$ (Theorem 19.1).]

10 A stronger identity suggested by the computations

The discovery phase suggested a compact formula not needed for any theorem above. We record it as a separate conjecture so that a promising observation is not silently promoted to a proof.

For an alphabet $X = (x_1, \dots, x_k)$, define $h_j(X)$ by $\sum_{j \geq 0} h_j(X) z^j = \prod_{i=1}^k (1 - x_i z)^{-1}$, with $h_j = 0$ for $j < 0$. The rectangular Schur polynomial is

$$s_{(n^a)}(X) = \det(h_{n-i+j}(X))_{1 \leq i, j \leq a}.$$

Conjecture 10.1 (Experimental rectangular-Schur formula). With $a = \lfloor k/2 \rfloor$ and $b = \lfloor (k-1)/2 \rfloor$,

$$\mathcal{H}_k(n; t) = s_{(n^a)}(\underbrace{1, \dots, 1}_a, t, \underbrace{t^2, \dots, t^2}_b). \quad (10.1)$$

The conjecture agrees with all 1,248 exact evaluations in the range recorded in Section 11.1. For instance, at $k = 4, n = 2$, its right-hand side is $h_2^2 - h_1 h_3$ in the alphabet $(1, 1, t, t^2)$, giving the polynomial displayed in Section 8.

A proof of (10.1) would give an attractive unified symmetric-function interpretation. It may be approachable through factorizations of rectangular Schur polynomials into classical-character alternants. This article does not assert such a factorization without proof. The proof of Conjecture 16 instead uses the parity factors, coefficientwise total nonnegativity, and the fully derived alternant formulas. The larger conjecture can be deleted from this article without changing any proved statement.

[Added 29 September 2026, batch 56: Conjecture 10.1 is proved in Part II for all $k \geq 1, n \geq 0$ as an identity in $\mathbb{Z}[t]$ (Theorem 12.1), through a Toeplitz factorization with a boundary parameter (Theorem 14.1). None of Part I's proofs is changed.]

11 Reproducibility and proof audit

11.1 What was checked computationally

The accompanying program `code/verify.py` uses exact integers and exact polynomials in $\mathbb{Z}[t]$. It does not infer symbolic identities from floating-point samples or from interpolation. The direct moment routine implements the source definition (1.1); the principal minor routine independently multiplies the banded matrices; and the small determinant routine implements (6.2).

Check	Range	Cases
Original Hankel versus parity formula	$1 \leq k \leq 12, 0 \leq n \leq 12$, eight integer parameters	1,248
Auxiliary corner versus alternant	$0 \leq a \leq 7, 0 \leq m \leq 8$, both factors, six parameters	864
Three-way polynomial equality; degree, positivity, reciprocity, stability, first defect	$1 \leq k \leq 10, 0 \leq n \leq 8$, in $\mathbb{Z}[t]$	90
Auxiliary first numerator term	$1 \leq a \leq 6, 0 \leq m \leq 6$, both numerators, in $\mathbb{Z}[t]$	84
Experimental Schur formula only	$1 \leq k \leq 12, 0 \leq n \leq 12$, eight integer parameters	1,248

The eight parameters are $-3, -2, -1, 0, 1, 2, 3, 5$. The six-parameter alternant test omits ± 1 , where its displayed denominator vanishes; these parameters are nevertheless covered by the original-versus-corner checks. All checks passed with Python 3.13.5 and SymPy 1.14.0. The execution record is in `data/verification.json`, with the console transcript in `data/verification.log` and coefficient arrays in `data/sample_polynomials.json`. A

supplementary script checks the specialization products in Appendix A in 234 cases and the recurrence in Proposition 9.1 in 126 cases; its exact ranges and results are recorded in `data/additional_verification.json`. These finite checks supplement the all-index proofs; they are not a substitute for them.

11.2 Logical dependencies and possible failure points

The main theorem depends only on the binomial definition, the Laurent basis calculation, elementary block row operations, Cauchy–Binet, monic orthogonal-polynomial recurrences, and alternating determinants. The proof of the moment-modification identity is included rather than used as an unexplained import. The exact role of each delicate point is as follows.

The odd/even row reorder is applied to both rows and columns, so there is no permutation sign. The only sign remaining is $(-1)^{\lfloor n/2 \rfloor}$ for odd shifts, which equals the prescribed $(-1)^{\binom{n}{2}}$. The temporary division by $1+t$ in the even-shift factorization is removed before specialization. The auxiliary normalization is justified by an identity in $\mathbb{Z}[t]$, not by dividing a determinant whose vanishing has not been established. Equal-variable alternants are evaluated after cancelling the Vandermonde, not by a 0/0 substitution. Finally, strict positivity is obtained from explicit Cauchy–Binet summands; nonnegativity alone would not have been sufficient.

11.3 Literature status and exclusions

The source pages and their mathematical notation were inspected directly. Searches by the paper identifier, its title, and the exact numbered conjecture did not locate a later proof of this specific statement. A search hit for Manuel Kauers's *A Proof of Conjecture 16* [4] was investigated separately: indexed material identifies its target as the sequence A339987, not Cigler's $c_n(t)$ conjecture. It was therefore not counted as a solution of the present problem. The full talk PDF was not successfully fetched, so this disambiguation rests on the indexed author material rather than a complete reading of the talk.

The article makes no absolute priority claim. It also does not claim to settle all parts of Cigler's Conjectures 17 and 18, or the separate Schur formula (10.1). What is claimed mathematically is the full argument for Conjecture 16 and the additional results explicitly proved here. Independent expert checking remains appropriate before publication or addition to a collection of established theorems.

[Added 29 September 2026, batch 56: Part II proves the Schur formula and corrected forms of the generating-function assertions of Conjecture 18; Conjecture 17 remains open (Section 21).]

A The removable specialization at $t = 1$

The factors have convenient product evaluations at 1:

$$\mathcal{U}_a(m; 1) = \prod_{r=1}^a \frac{2m+2r-1}{2r-1} \prod_{1 \leq r < s \leq a} \frac{2m+r+s-1}{r+s-1}, \quad (\text{A.1})$$

$$\mathcal{V}_a(m; 1) = \prod_{r=1}^a \frac{m+r}{r} \prod_{1 \leq r < s \leq a} \frac{2m+r+s}{r+s}. \quad (\text{A.2})$$

Empty products have value 1. To see these formulas directly from the proved alternants, set $x_i = e^{z_i}$ before taking the equal-parameter limit in (6.7). For $\mathcal{V}$, the numerator contains $\sinh((m+r)z_i)$ and the denominator $\sinh(rz_i)$, with $1 \leq r \leq a$. For $\mathcal{U}$, replace r by $r - \frac{1}{2}$. The

lowest nonzero alternating term of $\det(\sinh(\rho_j z_i))$ is a common nonzero constant times

$$\left(\prod_i z_i\right) \Delta(z_1^2, \dots, z_a^2) \left(\prod_j \rho_j\right) \Delta(\rho_1^2, \dots, \rho_a^2).$$

This follows by selecting the powers $1, 3, \dots, 2a - 1$ in the Taylor series. In the ratio, the differences between the ρ_j cancel, whereas the sums give the second products in (A.1)–(A.2). These specialize the new parity formula to the classical middle-binomial Hankel setting already treated in the source; the unweighted setting itself is not claimed as new.

B How to run the package

The archive contains the article source and PDF, three Python files, exact verification output, example coefficient arrays, and a status/source audit. The commands, from the archive's root directory, are

```
python -m pip install -r requirements.txt
python code/verify.py
python code/verify_additional.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The `Makefile` provides `make verify` and `make pdf`. The verification program stops at the first failed assertion or inexact division. It records the mathematical test ranges and software versions, not just a generic success message. Its stronger Schur checks are labelled `EXPERIMENTAL` in both the code and the output.

Part II. Rectangular Schur structure and sharp recurrences

Sections 1–11 and Appendices A and B form the original report (Part I) of 20 September 2026, unchanged except for dated pointers to this addition, marked [Added 29 September 2026, batch 56: ...]. Part II was added on 29 September 2026 in batch 56 of ProveIt’s incoming-report intake, from one manuscript written against the pinned state of Part I. It proves Part I’s experimental rectangular-Schur formula (Conjecture 10.1) and replaces the nonminimal recurrence of Proposition 9.1 by the exact minimal one. Its Sections 12–22 and Appendices C and D are the manuscript’s Sections 1–11 and Appendices A and B.

12 The selected question and the results

12.1 Provenance of Part II

Part II is batch-56 manuscript 01, *Rectangular Schur Structure and Sharp Recurrences for Cigler’s Hankel Polynomials: A proof of the repository’s Schur conjecture, exact pole multiplicities, and corrected generating-function reciprocity*, an AI-assisted research draft “prepared for Vladimir Reshetnikov” and dated 29 September 2026 (22 pages, A4, as delivered). It arrived as the archive `ProveIt_Cigler_Schur_Recurrences.zip` (inner directory `cigler_schur_research/`, main file `article.tex`) in ProveIt commit `e47ed8547` and was placed in `ae718a441`. It was written against ProveIt commit `e9c09b125` (in full `e9c09b12549ea4e3bea7cebc52b450b761f6f581`), at which Part I’s `article.tex` was the Git blob `f33543af1f` and the `article.tex` of the sibling report `cigler-conjecture-8-schur` [5] the blob `9c984baeeb`. No commit touched this report between `dc54c3cb3` (which brought the research-report collection into ProveIt) and the placement, so every statement the manuscript makes about “the repository’s report” or “the inspected ProveIt report” refers to Part I as printed above. The manuscript read Part I at the pinned commit through a GitHub connector, not from a library copy (its source audit, `02-schur-recurrences-sources.md`); Part I’s file is `SetTheory/Cardinals/docs/reports/hankel-determinants/catalan-and-ballot/cigler-conjecture-16-parity/article.tex`.

Its code and records are shipped beside Part I’s files with the prefix `02-schur-recurrences-` (Part I’s unprefixed files count as 01): the verifier `code/02-schur-recurrences-verify.py` and its Makefile `code/02-schur-recurrences-Makefile`; under `data/` the recorded run `02-schur-recurrences-verification.json` with its console log `02-schur-recurrences-verification.log`, the coefficient arrays `02-schur-recurrences-sample_polynomials.json`, the requirement pin `02-schur-recurrences-requirements.txt` and a hand-written layout record of the delivered PDF, `02-schur-recurrences-pdf_checks.json`; and in the report root its research-status record `02-schur-recurrences-STATUS.md` and its source audit `02-schur-recurrences-sources.md`. These files are byte-identical to the delivery, so their text still uses the delivery names. The manuscript, its PDF and its README are not shipped; they survive in the arrival commit. The report README maps every delivery name to its shipped path, and Appendix D explains why the shipped Makefile must not be run.

The manuscript contributes, in the order printed here: a proof of Part I’s Conjecture 10.1 for all $k \geq 1$ and $n \geq 0$ (Theorem 12.1), through a Toeplitz factorization with a boundary parameter (Theorem 14.1); a second derivation of Part I’s parity factorization (Proposition 13.1) and of Part I’s shape statements (Corollary 15.1); parity-wise unimodality for odd shifts (Proposition 15.2); a confluent spectral theorem for rectangular Schur polynomials with explicit nonzero leading coefficients (Theorem 16.1); the exact minimal recurrence of Cigler’s family (Theorem 12.2 and Corollary 17.1); the degree and reciprocity sign of the generating numerator (Lemma 18.1,

Theorem 18.2), with a correction of the sign printed in Cigler’s equation (85); the period-four signed normalization (Theorem 18.3); a sharp fixed-parameter convergence equivalent (Theorem 19.1); the reduced denominators at every real parameter (Theorem 19.2); its finite checks, proof-dependency audit and formalization plan (Section 20); six research topics (Section 21); a conclusion; and small-shift examples and package notes (Appendices C and D). Every result, proof, example, remark, limitation and question of the manuscript is printed.

Where the merge had to choose.

- *Duplicated material.* Section 13 re-derives Part I’s Sections 2–4 (Gram identities in the bases P_i , R_i and block elimination, in place of Part I’s tridiagonal principal minors); Corollary 15.1 recovers the shape statements of Theorem 1.1 through tableaux and is not claimed as new; equation (19.4) restates (1.6); Theorem 19.1 sharpens Proposition 9.2 for real $0 < t < 1$; the definitions (12.1)–(12.3) are those of (1.1), (1.2) and (1.3). The manuscript re-derives this material so that its argument does not rest on Part I; it is printed in both parts, and Part II’s proofs are second routes.
- *Numbering.* The manuscript’s Section n is Section $n + 11$ here, and its Theorem, Lemma, Proposition, Corollary and equation $n.m$ are $(n + 11).m$; its Appendices A and B are Appendices C and D. Its title block, abstract and status box are printed in Section 12.2, and this subsection and the next two precede its Section 1.1, which is Section 12.4 here. Its table of contents is merged into the report’s.
- *Names.* Every label of Part II carries the prefix **rs:**. The manuscript’s macro for $\mathcal{H}$ is replaced by Part I’s $\backslash\mathcal{H}$; both print the same glyph for the same object. Its bibliography key CK (Ciucu–Krattenthaler) is printed as CiucuK, because Part I’s CK is Cigler–Krattenthaler, a different paper; its entry for Part I is replaced by internal references, and its entry for Cigler’s paper is merged with Part I’s (the same version 3). No printed symbol is renamed and no normalization is changed. Symbols that mean different things in the two parts are listed in Section 12.3.
- *Added text.* Dated notes were added in Part I (title, abstract, Section 1.3, after Propositions 9.1 and 9.2, at the end of Section 10 and in Section 11.3), and four bracketed notes in Part II: on the pinned paths (Section 12.4), on Cigler’s printed page 24 (after (18.13); an observation of this merge, not of the manuscript), on the shipped names and a rerun (Section 20), and on the delivered commands (Appendix D). All are marked [Added 29 September 2026, batch 56: ...].

12.2 The manuscript’s title, abstract and status

The manuscript’s title block reads *Rectangular Schur Structure and Sharp Recurrences for Cigler’s Hankel Polynomials*, with the subtitle *A proof of the repository’s Schur conjecture, exact pole multiplicities, and corrected generating-function reciprocity*, the author line “Research draft prepared for Vladimir Reshetnikov” and the date 29 September 2026. Its abstract:

A report in the ProveIt repository leaves explicitly unproved a rectangular-Schur formula for normalized Hankel determinants of Cigler’s third polynomial extension of the middle binomial coefficients. We prove that formula for all shifts and orders. Writing $a = \lfloor k/2 \rfloor$ and $b = \lfloor (k - 1)/2 \rfloor$, the identity is

$$\mathcal{H}_k(n; t) = s_{(n^a)}((1)^{[a]}, t, (t^2)^{[b]}).$$

An elementary Toeplitz factorization supplies the missing bridge from the repository’s parity factors to the Schur determinant. We then prove a confluent spectral theorem for rectangular Schur polynomials: each allocation of rows among repeated-variable clusters has an explicit nonzero leading coefficient. For Cigler’s family this determines the *minimal* generic recurrence,

with characteristic polynomial

$$\prod_{q=0}^{k-1} (E - t^q)^{1 + \lfloor q(k-1-q)/2 \rfloor}.$$

We determine the exact generating-numerator degree and reciprocity sign, correct a sign in the printed Conjecture 18, obtain a sharp fixed-parameter convergence equivalent, and classify the reduced denominators at every real parameter. Exact finite checks and a focused further-research program accompany the all-index arguments.

Its status box:

Research status. The conclusions are mathematical proof claims, with the arguments supplied below. They have not been independently refereed or checked in Lean or Rocq. The unresolved repository formula is identified at a fixed commit; the general Schur, character, and confluence machinery is classical and is credited. No exhaustive historical-priority claim is made. The literal entirety of Cigler's printed Conjecture 18 is *not* asserted: one of its signs is false, and the corrected statements are given precisely.

12.3 Notation shared with Part I

Part II keeps the manuscript's notation. The two parts agree on $c_r(t)$, $D_k(n; t)$, $\mathcal{H}_k(n; t)$, a , b , $s = 1 + t$, u_r , v_r , the Laurent polynomial $Q(z) = Q_t(z) = (1 + tz)(1 + t/z)$, h_j , $s_{(n^a)}$ and the forward shift E . The following letters mean different things in the two parts; the right-hand column gives the tempting false reading where one exists.

Symbol	Part I	Part II	Do not read as
$\mathcal{U}, \mathcal{V}$	$\mathcal{U}_a(m; t) = \det(M_1^a[m])$, $\mathcal{V}_a(m; t) = \det(M_0^a[m])$: power a first, size m second	$\mathcal{U}_m(q)$, $\mathcal{V}_m(q)$: Gram determinants of size m of a symbol q , (13.1)–(13.2); $U_h(m) = \mathcal{U}_m(Q_t^h)$, $V_h(m) = \mathcal{V}_m(Q_t^h)$	$\mathcal{U}_m(q)$ as $\mathcal{U}_a(m; t)$ with $a = m$. The true dictionary is $\mathcal{U}_a(m; t) =$ $U_a(m) = \mathcal{U}_m(Q_t^a)$ and $\mathcal{V}_a(m; t) = V_a(m) =$ $\mathcal{V}_m(Q_t^a)$
italic U, V	$U_a(r)$, $V_a(r)$: unnormalized moment matrices in the proof of Theorem 4.1; U_a^∞ , V_a^∞ : limit series (7.2)	$U_h(m)$, $V_h(m)$: normalized determinants; the unnormalized matrices are U_r'	Part II's $U_h(m)$ as a matrix
$\mathcal{F}_k, F_k$	$\mathcal{F}_k(t)$ (1.4)	$F_k(t)$ (19.2), the same function	—
d	d_k : coarse exponent bound of Proposition 9.1; $d_k(n; t) = t^{-\binom{n}{2}} D_k(n; t)$	$d_k(n, t)$: the same unadjusted normalization (Cigler's); d_r , d_q , $d_\pm$: polynomial degrees in n	d_q as Part I's d_k
N	$N_{a,m}^\delta(t)$: numerator determinants (6.2)	N_k : order of the minimal recurrence (12.8)	—
δ	$\delta \in \{0, 1\}$ in $N_{a,m}^\delta$	$\delta_k = \lfloor k^2/4 \rfloor$ (18.1)	—
C	$C_j(y)$: Chebyshev-type polynomials in the proof of Theorem 6.2; $C_{k,r}$: constant of Proposition 9.2	$C_k = \binom{k}{2}$; $C = K - A$ in Section 16; $C_{2m}(u)$, $C_{2m+1}(u)$: normalizing scalars	—
B	$B = 1 + t^2$	$B(m, r)$: factorial ratio (16.3)	—
M	M_ε : tridiagonal matrices (2.4)	$M = k - 1$ (12.4)	—

Symbol	Part I	Part II	Do not read as
x, z	z : Laurent variable; x : generating variable of Proposition 9.1	z : Laurent variable (Sections 13–15) and generating variable (Sections 12 and 16–19, Appendix C); $x = z + z^{-1}$ in Sections 13–14	Cigler’s generating variable x (quoted in the note after (18.13)) as Part II’s $x = z + z^{-1}$
e	e_j : standard basis vectors	$e_{k,q}$: pole multiplicities; $e_j(X)$: elementary symmetric polynomials; (e_j) : bialternant exponents	—
h	— (Part I indexes the parity factors by a)	index of the parity factors in Section 13; derivative orders in Section 16	h as the complete homogeneous h_j
$A, \mathcal{A}$	$A[m]$: principal submatrix; A_j^ε : monic polynomials	A : height of the rectangle in Section 16; $\mathcal{A}_k$: generating numerator	—
J	$J = \text{diag}(1, -1, 1, \dots)$	the involution $Jf(z) = zf(z^{-1})$	—
L	the moment functional L	L_r : leading coefficients (16.5); LC: leading coefficient	—

12.4 Provenance and what is added

The source problem is Section 6 of Johann Cigler’s paper [1], version 3, dated 30 December 2021. Its moment family is denoted $c_n(t)$; it must not be confused with the paper’s other families $a_n(t)$ and $b_n(t)$. In ProveIt, the report *A parity factorization and sharp stabilization for Cigler’s Hankel polynomials* (Part I of this report) proves a parity factorization, coefficient positivity, reciprocity, stable edge coefficients, and a first-defect formula. Its Section 10 explicitly records an experimental rectangular-Schur identity as a conjecture, not a proof dependency. Its recurrence bound is explicitly nonminimal.

The inspected repository commit is

e9c09b12549ea4e3bea7cebc52b450b761f6f581.

The source article’s Git blob identifier is f33543af1f846c24511dc3004fb3542c2a20c50f. The selected conjecture belongs to the directory `cigler-conjecture-16-parity`; full paths are in the bibliography and the package’s source audit. The related repository report on *Conjecture 8* [5] uses a Laurent–Gram/Schur method for the *different* $b_n(t)$ moments. It is relevant methodological background, not a proof of the identity selected here.

[Added 29 September 2026, batch 56: in this report the bibliography entry for Part I is replaced by internal references; the pinned commit, blob identifiers and paths are recorded in Section 12.1, and the package’s source audit is shipped as `02-schur-recurrences-sources.md`.]

Our principal addition is the missing all-index Schur identity. Beyond that identity, the essential new calculation relative to the inspected reports is a noncancellation theorem for confluent spectral terms. It proves exact pole multiplicities, rather than merely giving a denominator that might cancel. The resulting numerator reciprocity also distinguishes a genuine statement from a sign error in the original conjecture.

Rectangular Schur factorizations at reciprocal alphabets are established mathematics; see Ciucu–Krattenthaler [6]. Recurrence existence for stretched Schur polynomials is also established; see Alexandersson [7]. Repeated-variable bialternants are treated explicitly by González-Serrano–Maximenko [8]. We use and explain these kinds of classical tools, rather than advertising

their existence as a discovery. The specific identity, exact multiplicities, constants, and source corrections are the objects proved here. A bounded literature search does not certify that no equivalent formulation has appeared elsewhere.

12.5 Definitions

For $r \geq 1$, put

$$c_r(t) = \sum_{j=1}^r \binom{\lfloor (r-1)/2 \rfloor}{\lfloor (j-1)/2 \rfloor} \binom{\lfloor r/2 \rfloor}{\lfloor j/2 \rfloor} t^{j-1}, \quad c_0(t) = 1. \quad (12.1)$$

A binomial coefficient is zero when its lower index is outside the usual range. These are Cigler's equation (6), with the zero $j = 0$ summand omitted. For a positive shift k and order $n \geq 0$, define

$$D_k(n; t) = \det(c_{k+i+j}(t))_{0 \leq i, j < n}, \quad (12.2)$$

$$\mathcal{H}_k(n; t) = (-1)^k \binom{n}{2} t^{-\binom{n}{2}} D_k(n; t). \quad (12.3)$$

Empty determinants are one. Initially (12.3) is read in $\mathbb{Q}(t)$; polynomiality and specialization at $t = 0$ are conclusions. Throughout, write

$$a = \lfloor k/2 \rfloor, \quad b = \lfloor (k-1)/2 \rfloor, \quad M = k-1, \quad \mathcal{X}_k = ((1)^{[a]}, t, (t^2)^{[b]}). \quad (12.4)$$

Here $x^{[r]}$ means r copies of x , including no copies when $r = 0$. The alphabet $\mathcal{X}_k$ has exactly k entries.

For an alphabet X , define $h_j(X)$ and $e_j(X)$ by

$$\sum_{j \geq 0} h_j(X) w^j = \prod_{x \in X} (1 - xw)^{-1}, \quad \sum_{j \geq 0} e_j(X) w^j = \prod_{x \in X} (1 + xw). \quad (12.5)$$

Subscripts below zero give zero. The Schur polynomial of the rectangle with a rows of length n has the classical Jacobi–Trudi descriptions

$$s_{(n^a)}(X) = \det(h_{n-i+j}(X))_{1 \leq i, j \leq a} = \det(e_{a-i+j}(X))_{1 \leq i, j \leq n}. \quad (12.6)$$

These identities, the tableau interpretation, and the bialternant formula are standard Schur identities [9, 8]. Empty rectangles have value one. They are the only symmetric-function identifications needed for the main determinant proof.

12.6 Main statements

Theorem 12.1 (The repository's rectangular-Schur conjecture). *For all $k \geq 1$ and $n \geq 0$, there is an identity in $\mathbb{Z}[t]$*

$$\boxed{D_k(n; t) = (-1)^k \binom{n}{2} t^{\binom{n}{2}} s_{(n^a)}(\mathcal{X}_k)}. \quad (12.7)$$

Consequently $\mathcal{H}_k(n; t) = s_{(n^a)}(\mathcal{X}_k)$, with no exceptional value of t .

Theorem 12.2 (Exact generic recurrence). *For $k \geq 1$, set*

$$e_{k,q} = 1 + \left\lfloor \frac{q(k-1-q)}{2} \right\rfloor, \quad N_k = \sum_{q=0}^{k-1} e_{k,q}. \quad (12.8)$$

Over $\mathbb{Q}(t)$, the reduced denominator of $\mathcal{G}_k(z, t) = \sum_{n \geq 0} \mathcal{H}_k(n; t) z^n$, normalized to have constant term one, is exactly

$$\boxed{\mathcal{D}_k(z, t) = \prod_{q=0}^{k-1} (1 - t^q z)^{e_{k,q}}}. \quad (12.9)$$

Equivalently, the monic minimal constant-coefficient recurrence operator is $\prod_q (E - t^q)^{e_{k,q}}$, where $Ef(n) = f(n+1)$. There is no cancellation in (12.9). The same statement holds at each real $t \notin \{0, 1, -1\}$.

The precise numerator and its sign are given in Theorem 18.2; the period-four signed normalization used by Cigler is treated in Theorem 18.3. Theorem 19.2 completes the real-parameter classification. The general mechanism that makes the recurrences minimal is Theorem 16.1, whose explicit leading coefficient also gives the sharp asymptotic equivalent in Theorem 19.1.

13 Laurent moments and the four parity factors

This section rederives the relevant part of Part I (Sections 2–4), so the new argument does not depend on trusting an unverified report.

13.1 Two elementary Gram identities

Let $q(z) = q(z^{-1})$ be a symmetric Laurent polynomial, and put $q_j = [z^j]q(z)$, so $q_{-j} = q_j$. Define

$$\mathcal{U}_m(q) = \det(q_{i-j} + q_{i+j+1})_{0 \leq i, j < m}, \quad (13.1)$$

$$\mathcal{V}_m(q) = \det(q_{i-j} - q_{i+j+2})_{0 \leq i, j < m}. \quad (13.2)$$

Both empty determinants are one. Set $x = z + z^{-1}$ and $y = z - z^{-1}$. For $i \geq 0$, the Laurent expressions

$$P_i(x) = \frac{z^{i+1} + z^{-i}}{z + 1}, \quad R_i(x) = \frac{z^{i+1} - z^{-i-1}}{z - z^{-1}} \quad (13.3)$$

are monic polynomials in x of degree i . Indeed, $P_0 = R_0 = 1$, $P_1 = x - 1$, $R_1 = x$, and each sequence satisfies $F_{i+1} = xF_i - F_{i-1}$. A direct multiplication gives

$$\frac{1}{2} \text{CT}_z q(z)(x+2)P_i(x)P_j(x) = q_{i-j} + q_{i+j+1}, \quad (13.4)$$

$$\frac{1}{2} \text{CT}_z q(z)(4-x^2)R_i(x)R_j(x) = q_{i-j} - q_{i+j+2}. \quad (13.5)$$

For example, $(x+2) = (z+1)^2/z$, and the product on the first line has four monomials before the constant term is taken. Symmetry of q merges them in pairs. For the second line use $4-x^2 = -y^2$. Thus $\mathcal{U}_m$ and $\mathcal{V}_m$ are ordinary polynomial Gram determinants for the two displayed constant-term weights. Monicity means that replacing P_i or R_i by x^i does not change either determinant.

13.2 The moments

Put

$$Q_t(z) = 1 + t^2 + t(z + z^{-1}) = (1 + tz)(1 + t/z), \quad s = 1 + t, \quad (13.6)$$

and define

$$u_r = \text{CT}_z (1+z)Q_t(z)^r, \quad v_r = \text{CT}_z (1-z^2)Q_t(z)^r. \quad (13.7)$$

Symmetry allows the alternative weights $(x+2)/2$ and $(4-x^2)/2$, respectively. Expansion of the two binomials in Q_t^r yields

$$u_r = \sum_{j=0}^r \binom{r}{j}^2 t^{2j} + \sum_{j=0}^{r-1} \binom{r}{j} \binom{r}{j+1} t^{2j+1}. \quad (13.8)$$

Separating parity in (12.1) and applying Pascal's identity gives

$$c_{2r+1} = u_r, \quad c_{2r+2} = su_r. \quad (13.9)$$

Moreover, $(s^2 - Q_t)(1 + z) = t(1 + z - z^2 - z^{-1})$. The constant terms of zQ_t^r and $z^{-1}Q_t^r$ agree, so

$$s^2 u_r - u_{r+1} = t v_r. \quad (13.10)$$

For short, write

$$U_h(m) = \mathcal{U}_m(Q_t^h), \quad V_h(m) = \mathcal{V}_m(Q_t^h) \quad (h, m \geq 0). \quad (13.11)$$

Since $Q_t = 1 + t^2 + tx$, its i th power has leading coefficient t^i as a polynomial in x . Applying the monic basis changes in (13.4)–(13.5) therefore gives

$$\det(u_{h+i+j})_{0 \leq i, j < m} = t^{m(m-1)} U_h(m), \quad (13.12)$$

$$\det(v_{h+i+j})_{0 \leq i, j < m} = t^{m(m-1)} V_h(m). \quad (13.13)$$

The exponent is twice $\sum_{i=0}^{m-1} i$. These are polynomial identities, not assertions obtained by substituting zero into an undefined quotient.

13.3 Block elimination

Proposition 13.1 (Parity formulas). *For $h, m \geq 0$,*

$$\mathcal{H}_{2h+1}(2m; t) = U_h(m) V_h(m), \quad (13.14)$$

$$\mathcal{H}_{2h+1}(2m+1; t) = U_h(m+1) V_h(m). \quad (13.15)$$

For $h \geq 1$ and $m \geq 0$,

$$\mathcal{H}_{2h}(2m; t) = U_h(m) V_{h-1}(m), \quad (13.16)$$

$$\mathcal{H}_{2h}(2m+1; t) = (1+t) U_{h-1}(m+1) V_h(m). \quad (13.17)$$

Proof. Reorder both rows and columns of the Hankel matrix with the even indices first. The permutation signs cancel. For an odd shift $2h+1$, the four blocks have entries u_{h+i+j} , su_{h+i+j} , su_{h+i+j} , and $u_{h+i+j+1}$. Subtract s times even row i from odd row i . The lower-left block becomes zero; the lower-right block becomes $-t(v_{h+i+j})$. Its size is m for both orders $2m$ and $2m+1$; the upper-left block has size m or $m+1$. Equations (13.12)–(13.13) give the normalization powers $2m^2 - m = \binom{2m}{2}$ and $2m^2 + m = \binom{2m+1}{2}$. The remaining sign $(-1)^m$ equals $(-1)^{\binom{n}{2}}$.

For an even shift $2h$ and even order, abbreviate the unnormalized moment matrices by $U'_r = (u_{r+i+j})_{0 \leq i, j < m}$. The reordered matrix is

$$\begin{pmatrix} sU'_{h-1} & U'_h \\ U'_h & sU'_h \end{pmatrix}.$$

Over $\mathbb{Q}(t)$, subtract s^{-1} times each lower row from the corresponding upper row. The upper-right block vanishes and the upper-left becomes $(t/s)(v_{h-1+i+j})$. The determinant is consequently $t^m \det(v_{h-1+i+j}) \det(u_{h+i+j})$.

For odd order $2m+1$, subtract s^{-1} times even row $i+1$ from odd row i , for $0 \leq i < m$. The lower-left block vanishes; the lower-right is $(t/s)(v_{h+i+j})$, and the upper-left is $s(u_{h-1+i+j})$ of size $m+1$. The remaining scalar factor is st^m . The same two normalization powers give (13.16) and (13.17). All final expressions are polynomials, so the temporary use of s^{-1} introduces no exception at $t = -1$. $\square$

14 The missing Toeplitz factorization

For a Laurent polynomial f , let

$$T_n(f) = \det([z^{j-i}]f(z))_{0 \leq i, j < n}. \quad (14.1)$$

The following two formulas are the bridge needed by the repository's experimental conjecture. We prove them by elementary finite-dimensional linear algebra, rather than requiring a classical-group character factorization as a black box.

Theorem 14.1 (Two factorizations with a boundary parameter). *For every symmetric Laurent polynomial q and every $m \geq 0$,*

$$T_{2m}((1+z)(1+u/z)q) = \mathcal{U}_m(Q_u q) \mathcal{V}_m(q), \quad (14.2)$$

$$T_{2m+1}((1+z)(1+u/z)q) = (1+u) \mathcal{U}_{m+1}(q) \mathcal{V}_m(Q_u q), \quad (14.3)$$

where $Q_u = 1 + u^2 + u(z + z^{-1})$. These are polynomial identities in u and the coefficients of q . In particular,

$$T_n((1+z)q) = \mathcal{U}_{\lceil n/2 \rceil}(q) \mathcal{V}_{\lfloor n/2 \rfloor}(q). \quad (14.4)$$

Proof. It is enough first to work over a characteristic-zero rational-function field, with $u \neq 0, 1, -1$ and independent coefficients for q . For Laurent polynomials define

$$\langle f, g \rangle_q = \text{CT}_z q(z) f(z) g(z^{-1}).$$

This form is symmetric. In consecutive monomial bases for a space V , the pairing of the left vectors $(1+z)z^i$ with the right vectors $(1+uz)z^j$ is the Toeplitz matrix of $(1+z)(1+u/z)q$. Translating all exponents by the same integer leaves its entries unchanged. We choose centered exponent ranges to expose the symmetry.

Even order. For $n = 2m$, take

$$V = \text{span}\{z^{-m}, \dots, z^{m-1}\}, \quad W = \text{span}\{z^{-m}, \dots, z^m\}.$$

Every element of W has a unique decomposition $p(x) + yr(x)$ with $\deg p \leq m$ and $\deg r \leq m-1$. The left space $(1+z)V$ is exactly the subspace of W vanishing at $z = -1$. Thus it has basis

$$(x+2)x^i \ (0 \leq i < m), \quad yx^i \ (0 \leq i < m).$$

Put

$$z_0 = -u^{-1}, \quad x_0 = -(u + u^{-1}), \quad y_0 = u - u^{-1}.$$

The right space $(1+uz)V$ is the kernel of evaluation at z_0 and has basis

$$(x - x_0)x^j \ (0 \leq j < m), \quad yx^j - y_0x_0^j \ (0 \leq j < m).$$

Linear independence follows separately from the symmetric and antisymmetric parts; their total dimensions are $2m$. Symmetric Laurent polynomials are orthogonal to antisymmetric ones for $\langle \cdot, \cdot \rangle_q$. The lower-left block is therefore zero. The two diagonal blocks are the moment matrices for the weights

$$(x+2)(x-x_0)q \quad \text{and} \quad (4-x^2)q.$$

Because $u(x-x_0) = Q_u$, their determinants are, up to factors depending only on u and m , $\mathcal{U}_m(Q_u q)$ and $\mathcal{V}_m(q)$ by (13.4)–(13.5).

Every basis change just made is independent of q . Consequently there is a nonzero rational function $C_{2m}(u)$, independent of q , such that the left side of (14.2) equals $C_{2m}(u) \mathcal{U}_m(Q_u q) \mathcal{V}_m(q)$. We determine this scalar below. This observation avoids hiding any powers of two, powers of u , or basis orientation signs.

Odd order. For $n = 2m + 1$, take

$$V = \text{span}\{z^{-m}, \dots, z^m\}, \quad W = \text{span}\{z^{-m}, \dots, z^{m+1}\}.$$

The involution $Jf(z) = zf(z^{-1})$ is an isometry of the pairing. Its two eigenspaces are orthogonal, and every element of W is uniquely

$$(1+z)p(x) + (z-1)r(x), \quad \deg p, \deg r \leq m.$$

A basis of the left space is

$$(1+z)x^i \ (0 \leq i \leq m), \quad (z-1)(x+2)x^i \ (0 \leq i < m).$$

This follows by writing the symmetric and antisymmetric parts of V and using $(1+z)y = (z-1)(x+2)$. Set $\gamma = (u-1)/(u+1)$. A basis of the right evaluation kernel is

$$(1+z)x^j + \gamma(z-1)x_0^j \ (0 \leq j \leq m), \quad (z-1)(x-x_0)x^j \ (0 \leq j < m).$$

Indeed $(1+z_0) + \gamma(z_0-1) = 0$, and dimensions and independence again follow from the two eigenspaces. The upper-right block is zero. The diagonal blocks now have weights

$$(x+2)q \quad \text{and} \quad (4-x^2)(x-x_0)q,$$

because $(z-1)(z^{-1}-1) = 2-x$. Their determinants are proportional to $\mathcal{U}_{m+1}(q)\mathcal{V}_m(Q_u q)$, with a scalar $C_{2m+1}(u)$ independent of q .

Normalization and specialization. For $q = 1$, both $\mathcal{U}_m(1)$ and $\mathcal{V}_m(1)$ equal one. Tridiagonal continuant recurrences give

$$\begin{aligned} T_n((1+z)(1+u/z)) &= 1 + u + \dots + u^n, \\ \mathcal{U}_m(Q_u) &= 1 + u + \dots + u^{2m}, \\ \mathcal{V}_m(Q_u) &= 1 + u^2 + \dots + u^{2m}. \end{aligned}$$

For example, all three recurrences follow by expanding the last row; the first has diagonal $1+u$ and off-diagonal product u , while the other two have diagonal $1+u^2$, off-diagonal product u^2 , and, for $\mathcal{U}$, an extra u in the first diagonal position. Their initial values fix the displayed sums. They imply $C_{2m} = 1$ and $C_{2m+1} = 1+u$. Both sides of the asserted identities are polynomials. Equality on the generic parameter field proves them identically, including $u = 0, 1, -1$ and singular choices of q . Substitution $u = 0$ gives (14.4). $\square$

15 Proof of the Schur identity and coefficient consequences

Proof of Theorem 12.1. Write $E_k(w) = \prod_{x \in \mathcal{X}_k} (1+xw)$ and, temporarily over $\mathbb{Q}(t)$, put

$$f_k(z) = t^a z^{-a} E_k(z/t). \quad (15.1)$$

Its coefficient at z^{j-i} is $t^{i-j} e_{a+j-i}(\mathcal{X}_k)$. The row factors t^i and column factors t^{-j} cancel in the determinant, so dual Jacobi–Trudi gives

$$T_n(f_k) = s_{(n^a)}(\mathcal{X}_k). \quad (15.2)$$

The point of the chosen alphabet is the following exact factorization:

$$f_k(z) = \begin{cases} (1+z)Q_t(z)^a, & k = 2a+1, \\ (1+z)(1+t/z)Q_t(z)^{a-1}, & k = 2a. \end{cases} \quad (15.3)$$

For instance, each paired factor contributes $tz^{-1}(1+z/t)(1+tz) = Q_t(z)$. For odd k , (14.4) yields exactly (13.14)–(13.15). For even k , use (14.2)–(14.3) with $u = t$ and $q = Q_t^{a-1}$; these yield (13.16)–(13.17). Thus $T_n(f_k) = \mathcal{H}_k(n; t)$ in all four cases. The resulting equality (12.7) is between polynomials in $\mathbb{Z}[t]$, so it holds there and at every specialization. The case $k = 1$ gives the empty-height Schur polynomial one. $\square$

15.1 A tableau model and reciprocal coefficients

The identity says that $\mathcal{H}_k(n; t)$ counts semistandard tableaux of shape (n^a) , with letters in an ordered alphabet of k symbols whose weights are 0 for the first a letters, 1 for the next, and 2 for the last b . The exponent of t is the sum of those weights. Thus the normalization in (12.3), including its sign, turns the original Hankel determinant into a genuine nonnegative generating polynomial.

Corollary 15.1 (Previously known shape properties, recovered). *The polynomial $\mathcal{H}_k(n; t)$ has exact degree $(k - 1)n$, constant and leading coefficients one, positive coefficients at every intermediate degree, and*

$$\mathcal{H}_k(n; t^{-1}) = t^{-(k-1)n} \mathcal{H}_k(n; t). \quad (15.4)$$

Proof. The all-zero-weight tableau is unique: every column must contain the first a letters in order. Thus the constant term is one. For odd k , the inverse alphabet is a permutation of $t^{-2}\mathcal{X}_k$. Homogeneity of Schur degree an proves (15.4). For even $k = 2a$, use the rectangular complement identity

$$s_{(n^{K-A})}(X) = \left(\prod X\right)^n s_{(n^A)}(X^{-1}) \quad (|X| = K). \quad (15.5)$$

It follows immediately by reversing the exponent columns in the bialternant, or by complementing columns of a rectangular tableau. Here $A = K - A = a$ and $\prod \mathcal{X}_k = t^{k-1}$. This proves reciprocity and the assertion about the top degree and coefficient.

For full support, regard a column as an increasing a -subset of $\{1, \dots, k\}$. There is a saturated componentwise chain from the first a letters to the last a letters, increasing a single letter by one at each step. Each step changes the column weight by zero or one, so the chain contains columns of all weights $0, 1, \dots, k - 1$. For a desired total weight between 0 and $(k - 1)n$, use some initial zero-weight columns, at most one intermediate-weight column from this chain, and some final maximum-weight columns. The columns are componentwise increasing, hence form a semistandard tableau. The empty cases are immediate. $\square$

These shape properties are not claimed as new relative to Part I (Theorem 1.1). The Schur identity gives a unified interpretation and an alternative proof. Ordinary unimodality is false: already

$$\mathcal{H}_4(2; t) = 1 + 2t + 5t^2 + 4t^3 + 5t^4 + 2t^5 + t^6.$$

A different, weaker coefficient inequality does follow for odd shifts.

Proposition 15.2 (Parity-wise unimodality for odd shifts). *If $k = 2a + 1$ and $\mathcal{H}_k(n; t) = \sum_j c_j t^j$, then $c_j = c_{2an-j}$ and $c_j \leq c_{j+2}$ whenever $0 \leq j \leq an - 1$. Thus the coefficients in each fixed parity class are unimodal.*

Proof. The eigenvalue alphabet $((t^{-1})^{[a]}, 1, (t)^{[a]})$ is the character alphabet of the $\mathrm{SL}_2(\mathbb{C})$ module $(\mathbb{C}^2)^{\oplus a} \oplus \mathbb{C}$. The classical Schur-module character theorem [9] identifies $t^{-an} \mathcal{H}_{2a+1}(n; t)$ with the character of the Schur functor of shape (n^a) on that module. Finite-dimensional SL_2 modules in characteristic zero are sums of irreducibles, whose characters are $t^r + t^{r-2} + \dots + t^{-r}$. Each such string is symmetric and has nondecreasing coefficients in steps of two toward the center. Their sum has the same properties. The representation-theoretic facts are imported classical results; they are not needed elsewhere in this article. $\square$

16 A confluent spectral theorem for rectangles

The next theorem separates recurrence existence from recurrence minimality. A determinant expansion easily produces exponential terms, but cancellation can remove their highest powers

of n . We identify a unique extremal confluent minor and evaluate its leading coefficient, which prevents that cancellation.

16.1 Statement with an explicit leading coefficient

Let $\xi_1, \dots, \xi_\ell$ be distinct nonzero elements of a field of characteristic zero. Give them positive multiplicities $m_1, \dots, m_\ell$, put $K = \sum_i m_i$, choose $1 \leq A < K$, and put $C = K - A$. For each allocation

$$0 \leq r_i \leq m_i, \quad \sum_i r_i = A, \quad (16.1)$$

define

$$s_i = m_i - r_i, \quad \lambda_r = \prod_i \xi_i^{r_i}, \quad d_r = \sum_i r_i s_i, \quad (16.2)$$

and use the factorial ratio

$$B(m, r) = \frac{\prod_{h=0}^{r-1} h!}{\prod_{h=m-r}^{m-1} h!}. \quad (16.3)$$

An empty product is one, so $B(m, 0) = B(m, m) = 1$.

Theorem 16.1 (Cluster allocation and exact polynomial degree). *There are polynomials $P_r(n)$ such that, for every $n \geq 0$,*

$$s_{(n^A)}((\xi_1)^{[m_1]}, \dots, (\xi_\ell)^{[m_\ell]}) = \sum_r P_r(n) \lambda_r^n. \quad (16.4)$$

The canonical polynomials obtained by the confluent Laplace expansion have exact degree d_r . Their leading coefficients are

$$L_r = (-1)^{\sum_{i < j} r_i s_j} \frac{\prod_i B(m_i, r_i) \xi_i^{C r_i - r_i s_i}}{\prod_{i < j} (\xi_j - \xi_i)^{s_i r_j + r_i s_j}}. \quad (16.5)$$

In particular every L_r is nonzero. If the products λ_r are pairwise distinct, the reduced generating function denominator is exactly

$$\prod_r (1 - \lambda_r z)^{d_r + 1}, \quad (16.6)$$

and the corresponding characteristic polynomial is the minimal recurrence.

Distinctness of the nodes is not the same as distinctness of the products. The first hypothesis suffices for the degree and leading-coefficient statements for the canonical summands. The second is essential to the last assertion: summands with a common product must otherwise be combined.

16.2 The confluent bialternant

For distinct variables $X = (x_1, \dots, x_K)$, the rectangular bialternant is

$$s_{(n^A)}(X) = \frac{\det(x_i^{e_j})_{1 \leq i, j \leq K}}{\prod_{i < j} (x_j - x_i)}, \quad (e_j) = (0, 1, \dots, C-1, n+C, \dots, n+K-1). \quad (16.7)$$

To coalesce m_i variables at ξ_i , replace their rows by the normalized derivative rows of orders $0, 1, \dots, m_i - 1$. Taylor expansion and multilinearity show that the within-cluster Vandermonde factor cancels between numerator and denominator. Thus the numerator matrix has rows (i, h) and entries

$$\binom{e_j}{h} \xi_i^{e_j - h}, \quad 0 \leq h < m_i, \quad (16.8)$$

and its denominator is

$$\Delta_m(\xi) = \prod_{i < j} (\xi_j - \xi_i)^{m_i m_j}. \quad (16.9)$$

Rows are ordered by increasing cluster label, then increasing derivative order. This convention fixes all signs. This limiting identity is a standard repeated-variable bialternant; [8] supplies general treatments. The Taylor argument just described explains the factorial normalization used here.

Expand the numerator along its last A columns. Suppose these columns select r_i rows from cluster i , with derivative-order set R_i . The remaining C columns contain no n . From every selected cluster row in the last block one can extract ξ_i^n , leaving a polynomial in n . Grouping by $(r_1, \dots, r_\ell)$ proves the existence of (16.4). The main issue is its exact degree.

16.3 Degree loss forced by confluent rows

Write $D^h f = f^{(h)}/h!$ and $g_j(x) = x^{C+j}$ for $0 \leq j < A$. Normalized Leibniz expansion gives

$$D^h(x^n g_j(x))|_{x=\xi_i} = \xi_i^n \sum_{p=0}^h \binom{n}{h-p} \xi_i^{p-h} D^p g_j(\xi_i). \quad (16.10)$$

For a fixed cluster, the r_i selected rows are linear combinations of the rows $(D^p g_j(\xi_i))_j$. In a nonzero determinant contribution, the selected p 's in that cluster must be distinct. Cauchy–Binet makes this statement literal: the coefficient matrix is block diagonal by cluster, so each nonzero minor selects exactly r_i derivative rows from cluster i . The sum of their orders is at least $\binom{r_i}{2}$. Hence the degree of the high-column determinant, after removing λ_r^n , is at most

$$\sum_i \left(\sum_{h \in R_i} h - \binom{r_i}{2} \right). \quad (16.11)$$

Among all r_i -subsets of $\{0, \dots, m_i - 1\}$, the sum of orders is uniquely maximized by

$$R_i = \{s_i, s_i + 1, \dots, m_i - 1\}. \quad (16.12)$$

For that set the contribution to (16.11) is $r_i s_i$. Every other row selection has strictly lower total degree. This proves that only one Laplace row selection can contribute to n^{dr} . It remains to show that its coefficient is nonzero and to calculate it.

16.4 Evaluation of the unique leading term

For the choice (16.12), the lowest possible derivative orders in (16.10) are $p = 0, \dots, r_i - 1$. Since $\binom{n}{h-p}$ has leading coefficient $1/(h-p)!$, the local leading coefficient determinant is

$$\det \left(\frac{1}{(s_i + h - p)!} \right)_{0 \leq h, p < r_i} = B(m_i, r_i), \quad (16.13)$$

where a negative factorial is interpreted as giving entry zero. To verify this identity, multiply row h by $(s_i + h)!$. The entry in column p becomes the monic falling-factorial polynomial $(s_i + h)_p$ of degree p . Its determinant is the Vandermonde of $s_i, s_i + 1, \dots, s_i + r_i - 1$, namely $\prod_{h=0}^{r_i-1} h!$. Division by the row factors proves (16.13). The powers of ξ_i in (16.10) supply $\xi_i^{-r_i s_i}$.

The low-column minor uses derivative orders $0, \dots, s_i - 1$ in each cluster, so its determinant is $\prod_{i < j} (\xi_j - \xi_i)^{s_i s_j}$. The high-column derivative matrix with orders $0, \dots, r_i - 1$ is the confluent

matrix of $g_j(x) = x^C x^j$. Multiplication by x^C induces, within each cluster, a lower-triangular derivative change with diagonal ξ_i^C . Its determinant is therefore

$$\left(\prod_i \xi_i^{Cr_i} \right) \prod_{i < j} (\xi_j - \xi_i)^{r_i r_j}.$$

In the full Laplace expansion the selected high rows follow the low rows within each cluster. Moving all low rows before all high rows crosses $r_i s_j$ pairs for each $i < j$, so the sign is $(-1)^{\sum_{i < j} r_i s_j}$. Finally, divide by (16.9). The cross exponent becomes $m_i m_j - s_i s_j - r_i r_j = s_i r_j + r_i s_j$, giving exactly (16.5). None of its factors vanishes. This proves the exact degree.

16.5 Why the recurrence is minimal

For a nonzero polynomial P of degree d and nonzero λ ,

$$(E - \lambda)(P(n)\lambda^n) = \lambda^{n+1}(P(n+1) - P(n)).$$

Thus $(E - \lambda)^{d+1}$ kills this sequence, whereas the d th power does not. If $\mu \neq \lambda$, applying $E - \mu$ preserves its polynomial degree and multiplies its leading coefficient by $\lambda - \mu$. These observations prove independence of the spaces $\{P(n)\lambda^n\}$ for distinct λ : apply operators killing all but one of a putative relation. Equivalently, $\sum_{n \geq 0} P(n)(\lambda z)^n$ has a pole of exact order $d+1$ at $z = \lambda^{-1}$; its top pole coefficient is a nonzero multiple of $d! \text{LC}(P)$. Distinct poles cannot cancel one another. Applying this to (16.4) proves (16.6) and finishes Theorem 16.1.

17 Exact recurrences for Cigler's family

Proof of Theorem 12.2. For $k \geq 2$, specialize Theorem 16.1 to cluster nodes $(1, t, t^2)$ and multiplicities $(a, 1, b)$, omitting a zero-multiplicity cluster. The height is $A = a$. An allocation satisfies $r_1 + r_2 + r_3 = a$ and $r_2 \in \{0, 1\}$. Its product is t^q , where $q = r_2 + 2r_3$. For each feasible q the allocation is unique:

$$r_3 = \lfloor q/2 \rfloor, \quad r_2 = q - 2\lfloor q/2 \rfloor, \quad r_1 = a - \lfloor q/2 \rfloor. \quad (17.1)$$

The feasible values are exactly $0 \leq q \leq k-1$. Uniqueness is crucial: the algebraic relation between the three nodes creates no collision between different allocations because the middle cluster has multiplicity one.

The exact degree is

$$d_q = r_1(a - r_1) + r_3(b - r_3) = \left\lfloor \frac{q(k-1-q)}{2} \right\rfloor. \quad (17.2)$$

For example, if $q = 2h$ the first expression is $h(a + b - 2h)$. If $q = 2h + 1$, it is $(a - h - 1)(h + 1) + h(b - h)$; substituting $b = a$ or $a - 1$ gives the displayed floor in either parity of k . Thus

$$\mathcal{H}_k(n; t) = \sum_{q=0}^{k-1} P_{k,q}(n; t) t^{q^n}, \quad \deg_n P_{k,q} = e_{k,q} - 1, \quad (17.3)$$

with nonzero leading coefficients supplied by (16.5). Over $\mathbb{Q}(t)$ the products t^q are distinct; Theorem 16.1 proves minimality. At a real parameter outside $\{0, 1, -1\}$, the nodes $1, t, t^2$ and the products t^q remain distinct and nonzero, and all the same leading coefficients remain nonzero. The case $k = 1$ is simply the constant sequence one. $\square$

Corollary 17.1 (Exact recurrence orders). *For $a \geq 1$,*

$$N_{2a} = 2a + \frac{a(a-1)(2a-1)}{3}, \quad (17.4)$$

$$N_{2a+1} = 2a + 1 + \frac{2a(a^2-1)}{3}. \quad (17.5)$$

Also $N_1 = 1$.

Proof. Sum (12.8). If $k = 2a$, the product $q(2a-1-q)$ is always even, so there is no rounding. If $k = 2a+1$, subtract $1/2$ for each odd q from $q(2a-q)/2$; there are a such values. The usual sums of integers and squares give the formulas. $\square$

k	Pole multiplicities $(e_{k,0}, \dots, e_{k,k-1})$	N_k	$N_k - k$
1	(1)	1	0
2	(1, 1)	2	0
3	(1, 1, 1)	3	0
4	(1, 2, 2, 1)	6	2
5	(1, 2, 3, 2, 1)	9	4
6	(1, 3, 4, 4, 3, 1)	16	10
7	(1, 3, 5, 5, 5, 3, 1)	23	16
8	(1, 4, 6, 7, 7, 6, 4, 1)	36	28

This is stronger than recurrence existence and stronger than the repository's coarse product of powers of $E^2 - t^{2q}$. The proof identifies every needed root and every needed multiplicity. It also explains why a numerical fit at $t = 1$ would be misleading: that parameter merges all the roots, producing a very different reduced recurrence.

17.1 Exact computation without a large Hankel matrix

For fixed k , Jacobi–Trudi computes $\mathcal{H}_k(n; t)$ as an $a \times a$ determinant, not an $n \times n$ determinant. Its entries are generated by

$$\sum_{j \geq 0} h_j(\mathcal{X}_k) w^j = \frac{1}{(1-w)^a (1-tw)(1-t^2w)^b}. \quad (17.6)$$

For many successive orders, one can instead compute an initial segment and use the exact order- N_k recurrence. For an isolated very large order, recurrence exponentiation gives an alternative whose field-operation count is logarithmic in n for fixed k . None of these statements removes output size or integer bit complexity: an expanded polynomial still has degree $(k-1)n$, and integer specializations may have linearly growing bit length. The polynomial $P_{k,q}(n; t)$ can be obtained directly from the finite confluent Laplace expansion, without guessing it by interpolation.

18 Generating numerators, negative indices, and a corrected sign

Set

$$\delta_k = a(k-a) = \lfloor k^2/4 \rfloor, \quad C_k = \binom{k}{2}, \quad R_k = N_k - k. \quad (18.1)$$

Here δ_k is not a Hankel determinant. It is the degree of the dimension polynomial that appears at $t = 1$.

18.1 Negative-width continuation

The exponential-polynomial expression (17.3) defines $\mathcal{H}_k(n; t)$ for every integer n , over $\mathbb{Q}(t)$. This is an algebraic continuation in the width index, not a combinatorial interpretation of a negative rectangle.

Lemma 18.1 (A zero gap and negative-index reciprocity). *For $k \geq 2$,*

$$\mathcal{H}_k(-j; t) = 0 \quad (1 \leq j \leq k-1), \quad (18.2)$$

$$\mathcal{H}_k(-k-r; t) = (-1)^{\delta_k} t^{-C_k-(k-1)r} \mathcal{H}_k(r; t) \quad (r \geq 0). \quad (18.3)$$

For $k = 1$, the second identity holds and the zero range is empty.

Proof. Use the bialternant (16.7), initially for distinct nonzero variables. Its exponent columns are $0, \dots, K-A-1$ and $n+K-A, \dots, n+K-1$. If $n = -j$ with $1 \leq j \leq K-1$, the two intervals overlap, so two columns coincide. If $n = -K-r$, the second interval is $-A-r, \dots, -r-1$. Move that block before the first block, with sign $(-1)^{A(K-A)}$, then multiply each variable row by x_i^{A+r} . The resulting bialternant is that of $s_{(rK-A)}(X)$. Therefore the continued value is

$$(-1)^{A(K-A)} (\prod X)^{-A-r} s_{(rK-A)}(X) = (-1)^{A(K-A)} (\prod X)^{-A} s_{(rA)}(X^{-1}),$$

where the last step is (15.5). The identities survive confluence. On $\mathcal{X}_k$, one has $(\prod \mathcal{X}_k)^a = t^{C_k}$, and (15.4) gives the remaining factor. Since both the bialternant and the spectral expansion agree for all $n \geq 0$, their polynomial-exponential extensions agree identically. $\square$

18.2 The exact numerator theorem

Theorem 18.2 (Degree and corrected reciprocity). *The reduced generating function has the form*

$$\mathcal{G}_k(z, t) = \frac{\mathcal{A}_k(z, t)}{\mathcal{Q}_k(z, t)}, \quad \mathcal{A}_k \in \mathbb{Z}[t, z], \quad \mathcal{A}_k(0, t) = 1. \quad (18.4)$$

Its numerator has exact z -degree $R_k = N_k - k$, with leading coefficient

$$[z^{R_k}] \mathcal{A}_k = (-1)^{N_k + \delta_k + 1} t^{(k-1)R_k/2}, \quad (18.5)$$

and satisfies

$$t^{(k-1)R_k/2} z^{R_k} \mathcal{A}_k\left(\frac{1}{t^{k-1}z}, t\right) = (-1)^{N_k + \delta_k + 1} \mathcal{A}_k(z, t). \quad (18.6)$$

The sign is $(-1)^{a+1}$ when $k = 2a$ and is $+1$ when k is odd.

Proof. Every exponential-polynomial sequence $F(n)$ has the following expansion of its ordinary rational generating function at infinity:

$$\sum_{n \geq 0} F(n) z^n = - \sum_{j \geq 1} F(-j) z^{-j}. \quad (18.7)$$

This is a statement about expansions of one rational function at two places, not an equality between two convergent series on a common annulus. For $F(n) = \lambda^n$ it is the two expansions of $(1 - \lambda z)^{-1}$. Applying $(z d/dz)^h$ proves it for $n^h \lambda^n$, hence in general.

Lemma 18.1 shows that the first nonzero term at infinity is $-\mathcal{H}_k(-k; t) z^{-k} = (-1)^{\delta_k + 1} t^{-C_k} z^{-k}$. The denominator has degree N_k , so the numerator has degree $N_k - k$. Its leading coefficient is obtained by multiplying that term by the leading coefficient $(-1)^{N_k} t^{\sum_q q e_{k,q}}$ of $\mathcal{Q}_k$. Symmetry

$e_{k,q} = e_{k,k-1-q}$ gives $\sum_q q e_{k,q} = (k-1)N_k/2$, proving (18.5). The recurrence makes the numerator a finite convolution of polynomials in $\mathbb{Z}[t]$, so it lies in $\mathbb{Z}[t, z]$.

Substitute $1/(t^{k-1}z)$ in the expansion at infinity and use (18.3). Since $k(k-1) = 2C_k$, this gives

$$\mathcal{G}_k\left(\frac{1}{t^{k-1}z}, t\right) = (-1)^{\delta_k+1} t^{C_k} z^k \mathcal{G}_k(z, t). \quad (18.8)$$

On the other hand,

$$\mathcal{Q}_k\left(\frac{1}{t^{k-1}z}, t\right) = (-1)^{N_k} t^{-(k-1)N_k/2} z^{-N_k} \mathcal{Q}_k(z, t).$$

Combining these two formulas proves (18.6). For even $k = 2a$, N_k is even and $\delta_k = a^2$. For odd k , N_k is odd and $\delta_k = a(a+1)$ is even. These parity facts give the stated signs. They also show explicitly that all displayed half-exponents of t are integers. $\square$

18.3 What this says about the printed Conjecture 18

For even shift $k = 2a$, Cigler's $d_{2a}(n, t)$ is exactly our $\mathcal{H}_{2a}(n; t)$. The denominator exponent in his equation (84) is

$$aj - \frac{(j-1)(j+2)}{2} = 1 + \frac{j(2a-1-j)}{2} = e_{2a,j}.$$

Thus the present theorem proves that proposed denominator and proves it is minimal. Its numerator degree is

$$R_{2a} = \frac{a(a-1)(2a-1)}{3}.$$

However, the positive-sign reciprocity printed in equation (85) is false when a is even. The corrected sign is $(-1)^{a+1}$. This is not a subtle large-index discrepancy: the same source prints

$$\mathcal{G}_4(z, t) = \frac{1 - t^3 z^2}{(1-z)(1-tz)^2(1-t^2 z^2)(1-t^3 z)}. \quad (18.9)$$

For its numerator, $t^3 z^2 \mathcal{A}_4(1/(t^3 z), t) = t^3 z^2 - 1 = -\mathcal{A}_4(z, t)$. This directly contradicts the positive sign in the printed equation. We therefore prove a corrected formulation, not the literal whole of Conjecture 18. The original source, printed page 24, was inspected directly for this comparison [1].

18.4 The period-four signed normalization

The source uses $d_k(n, t) = t^{-\binom{n}{2}} D_k(n; t)$. For odd k this is $\sigma_n \mathcal{H}_k(n; t)$, where

$$\sigma_n = (-1)^{\binom{n}{2}} = \frac{1-i}{2} i^n + \frac{1+i}{2} (-i)^n. \quad (18.10)$$

The two normalizations must not be interchanged in a generating function.

Theorem 18.3 (Exact signed denominator and numerator). *Let $k = 2a + 1$ and write $\widehat{\mathcal{G}}_k(z, t) = \sum_{n \geq 0} d_k(n, t) z^n$. Its reduced denominator over $\mathbb{Q}(t)$ is exactly*

$$\widehat{\mathcal{D}}_k(z, t) = \prod_{q=0}^{k-1} (1 + t^{2q} z^2)^{e_{k,q}}. \quad (18.11)$$

The numerator $\widehat{\mathcal{A}}_k$ lies in $\mathbb{Z}[t, z]$, has constant term one and exact degree $\widehat{R}_k = 2N_k - k$, and satisfies

$$t^{(k-1)\widehat{R}_k/2} z^{\widehat{R}_k} \widehat{\mathcal{A}}_k\left(\frac{1}{t^{k-1}z}, t\right) = (-1)^a \widehat{\mathcal{A}}_k(z, t). \quad (18.12)$$

Proof. Equation (18.10) changes each spectral root t^q into the two roots it^q and $-it^q$, with nonzero coefficients. They are all distinct over $\mathbb{Q}(t, i)$, so the minimality argument doubles the recurrence order and gives (18.11). The product has rational coefficients, and minimality over the extension implies minimality over $\mathbb{Q}(t)$. For negative indices, $\sigma_{-k-r} = (-1)^{a+1}\sigma_r$. Since δ_k is even, the proof of (18.8) becomes

$$\widehat{\mathcal{G}}_k\left(\frac{1}{t^{k-1}z}, t\right) = (-1)^a t^{C_k} z^k \widehat{\mathcal{G}}_k(z, t).$$

The gap of $k-1$ zero negative indices gives degree $2N_k - k$ as before. The quadratic denominator transforms without a sign, with factor $t^{-(k-1)N_k} z^{-2N_k}$. This proves (18.12) and the remaining assertions. $\square$

As a useful normalization check, the first nontrivial odd shift gives

$$\widehat{\mathcal{G}}_3(z, t) = \frac{1 + (1+t+t^2)z - t(1+t+t^2)z^2 - t^3z^3}{(1+z^2)(1+t^2z^2)(1+t^4z^2)}. \quad (18.13)$$

It has $d_3(1, t) = c_3(t) = 1 + t + t^2$, as the moment definition requires. The cubic term and the sign in (18.13) cannot be omitted. This provides a second direct check on the printed small-shift generating functions, independent of the general numerator proof.

[Added 29 September 2026, batch 56. Observation of this merge, not of the manuscript; it concerns the external source only. The generating function for d_3 printed on p. 24 of [1], with numerator $1 + (1+t)^2x + t^2x^2$ over $(1+x^2)(1+t^2x^2)(1+t^4x^2)$, is incorrect: its coefficient of x is $(1+t)^2$, whereas $d_3(1, t) = c_3(t) = 1 + t + t^2$ (at $t = 2$: 9 against 7). The numerator of (18.13) factors as $(1-tz)(1+(1+t)^2z+t^2z^2)$, so the printed numerator omits the factor $1-tx$. Equation (18.13) agrees with the determinants for $n < 10$ at $t = 2, 3, -5/2$, and so do the printed generating functions for d_1, d_2 and d_4 . In the same source, equation (87), for the shift $2k+1$, has the sign $(-1)^k$ and the argument $1/(t^{2k}x)$ of Theorem 18.3 with $a = k$, but its exponents of x and t , $\rho(k) = \frac{2k(k-1)(2k-1)}{3} - 2(k-1)^2 + 1$ and $k\rho(k)$, hold only with $\rho(k+1) = (4k^3 + 2k + 3)/3 = \widehat{R}_{2k+1}$ in place of $\rho(k)$: for the shift 3 the printed exponent is $\rho(1) = 1$, while the numerator (18.13) has degree 3. So (87) mixes the indices of two shifts; its sign is right. Both readings were checked on a rendering of the printed page of arXiv:2111.14492v3 and by exact computation during the intake.]

19 Sharp convergence and all real specializations

19.1 The leading nonconstant spectral term

Theorem 19.1 (Sharp fixed-parameter convergence equivalent). *Fix $k \geq 2$ and $0 < t < 1$. Then, as $n \rightarrow \infty$,*

$$\frac{1}{(1-t)^a(1-t^2)^{ab}} - \mathcal{H}_k(n; t) \sim \frac{n^{a-1}t^{n+1}}{(a-1)!(1-t)^{b+1}(1-t^2)^{(a-1)b}}. \quad (19.1)$$

The equivalent is uniform for t in each compact subinterval of $(0, 1)$; more precisely, its relative error is $O(1/n)$ on such a subinterval.

Proof. At $q = 0$, the allocation is $(a, 0, 0)$ and $d_0 = 0$. Formula (16.5) consequently gives the whole constant spectral term, not merely a leading approximation:

$$P_{k,0}(n; t) = F_k(t) := \frac{1}{(1-t)^a(1-t^2)^{ab}}. \quad (19.2)$$

At $q = 1$, the allocation is $(a - 1, 1, 0)$, of degree $a - 1$. Here $B(a, a - 1) = 1/(a - 1)!$. Substitution into (16.5), followed by cancellation of the factors $t^2 - t = t(t - 1)$, yields

$$[n^{a-1}]P_{k,1}(n; t) = -\frac{t}{(a-1)!(1-t)^{b+1}(1-t^2)^{(a-1)b}}. \quad (19.3)$$

All remaining terms in (17.3) contain t^{qn} with $q \geq 2$. For fixed k , their polynomial degrees are bounded and their rational coefficients are bounded on any compact subinterval of $(0, 1)$. They are exponentially smaller than $n^{a-1}t^n$ there. The lower coefficients of $P_{k,1}$ give relative error $O(1/n)$; when $a = 1$ there are no such coefficients. Equations (19.2)–(19.3) prove the assertion. $\square$

The repository already identifies F_k as the limiting series and proves the formal first defect

$$\mathcal{H}_k(n; t) = F_k(t) - \binom{n+a}{a-1} t^{n+1} + O(t^{n+2}) \quad \text{in } \mathbb{Q}[[t]] \quad (19.4)$$

for fixed n (Part I, Theorem 1.1). The limit in (19.1) is a different statement: t is fixed and n grows, so all powers beyond t^{n+1} can affect the prefactor. The rational factor in (19.1) accounts for those contributions. In particular, a formal first-defect formula alone would not justify this asymptotic equivalent. For example,

$$F_4(1/2) - \mathcal{H}_4(n; 1/2) \sim \frac{8}{3} n 2^{-n}.$$

No uniform assertion near $t = 1$ is made; the pole collision there requires a different scaling analysis.

19.2 The exceptional real parameters

Theorem 19.2 (Complete real-parameter denominator classification). *For $\sum_{n \geq 0} \mathcal{H}_k(n; t)z^n$, the reduced denominator at every real parameter is as follows, always normalized to constant term one:*

parameter	reduced denominator
$t \notin \{0, 1, -1\}$	$\prod_{q=0}^{k-1} (1 - t^q z)^{e_{k,q}}$
$t = 0$	$1 - z$
$t = 1$	$(1 - z)^{\delta_k + 1}$
$t = -1, k = 2a$	$(1 - z^2)^{a(a-1)+1}$
$t = -1, k = 2a + 1$	$(1 - z)^{a^2+1}(1 + z)^{a^2}.$

(19.5)

All exponents are exact, not just upper bounds. In particular the last line includes $k = 1$ with $a = 0$.

Proof. The generic real case was proved in Theorem 12.2. At zero, the alphabet contains a ones and zeros. Only the tableau using all a one-valued letters in each column contributes, so $\mathcal{H}_k(n; 0) = 1$.

At one, every alphabet entry is one. The confluent bialternant reduces to the elementary dimension product

$$\mathcal{H}_k(n; 1) = \prod_{i=1}^a \prod_{j=a+1}^k \frac{n+j-i}{j-i}. \quad (19.6)$$

One can derive this directly: the determinant of the normalized derivative matrix at one is the Vandermonde of its exponent columns divided by $\prod_{h=0}^{k-1} h!$. The within-block factors cancel,

leaving precisely the cross-block ratios in (19.6). The product is a nonzero polynomial in n of exact degree δ_k . Its generating denominator is therefore $(1 - z)^{\delta_k + 1}$.

At minus one, the alphabet is 1 repeated $k - 1$ times together with a single -1 . For $k \geq 2$, apply Theorem 16.1 with these two nodes. There are only two allocations: the -1 cluster contributes zero or one selected row. The spectral roots are 1 and -1 , and the polynomial degrees are respectively

$$d_+ = a(k - 1 - a) = ab, \quad d_- = (a - 1)(k - a) = (a - 1)(b + 1).$$

Their leading coefficients are nonzero by (16.5). If $k = 2a$, both degrees are $a(a - 1)$; if $k = 2a + 1$, they are a^2 and $a^2 - 1$. This proves the two last lines. For $k = 1$ the sequence is one directly. $\square$

The determinant values at $t = 1$ and the Catalan-related simplifications at $t = -1$ are related to the source's specializations ([1]; Part I, Section 8 and Appendix A). The conclusion here is an exact reduced-recurrence classification at those specializations, including the proof that the surviving poles really occur. It would be incorrect to substitute $t = 1$ into the generic product and call the resulting multiplicity N_k minimal: the actual multiplicity is $\delta_k + 1$.

20 Reproducibility, proof dependencies, and formalization

20.1 What the accompanying program actually checks

The package contains `code/verify.py`, a runnable verifier. It uses fraction-free Bareiss elimination with an exact-division check at every step. The original binomial moments and the Schur side are constructed independently. For the latter, complete homogeneous polynomials are computed from (17.6), then an $a \times a$ Jacobi–Trudi determinant is evaluated. All numerical checks use integers or rational numbers. SymPy is used only for exact polynomial-ring checks.

The run supplied in `data/verification.json` and `data/verification.log` used Python 3.13.5 and SymPy 1.14.0. All of the following checks passed:

Check	Range or method	Cases
Original Hankel vs. Schur	$1 \leq k \leq 12, 0 \leq n \leq 12; t = -3, -2, -1, 1, 2, 3, 5$	1,092
Universal Toeplitz factorization	12 signed symmetric symbols; $0 \leq n \leq 12; u = -3, -2, -1, 0, 1, 2, 3$	1,092
Ordinary recurrence and numerator	$1 \leq k \leq 12; t = -2, 2, 3$; sequence checked through $2N_k + 8$	36
Period-four signed recurrence	Odd k from 1 to 11; $t = -2, 2, 3$	18
Numerator reciprocity coefficients	All coefficients of both tested numerator families	2,052
Confluent leading coefficients	Five cluster-multiplicity families at prime nodes; also Cigler $2 \leq k \leq 10, t = 2$	143
Exceptional real parameters	$1 \leq k \leq 12, t = 0, 1, -1$	36
Exact polynomial Hankel–Schur identity	$1 \leq k \leq 8, 0 \leq n \leq 6$ in $\mathbb{Z}[t]$	56

The polynomial checks also test reciprocity, positive support, and the odd-shift parity-wise coefficient inequalities. Their actual coefficient arrays are saved in `data/sample_polynomials.json`. The universal Toeplitz test uses a reproducible pseudorandom seed, 260929, and includes negative coefficients, so it is not restricted to positive moment forms. The JSON record lists every cluster family and range explicitly.

[Added 29 September 2026, batch 56: in this report the verifier is shipped as `code/02-schur-recurrences-verify.py` and its records as `data/02-schur-recurrences-verification`.

json, data/02-schur-recurrences-verification.log and data/02-schur-recurrences-sample_polynomials.json; code/verify.py and data/verification.json are Part I's files (Section 11.1). Run the verifier with an explicit `-output` outside `data/`: by default it writes `data/verification.json`, and it always writes `sample_polynomials.json` next to its output, so a run without `-output` would overwrite Part I's records. A rerun on a copy on 29 September 2026 (Python 3.13.5, SymPy 1.14.0) passed every check and reproduced the recorded files up to line endings (the script writes CRLF on Windows) and the field `elapsed_seconds`. For the exceptional parameters the recorded run checks that the denominators of Theorem 19.2 annihilate the sequences; their minimality is proved in the text, and an independent Berlekamp–Massey computation during the intake found exactly those minimal denominators for $k \leq 10$ at $t = 0, \pm 1$, and exactly $\mathcal{Q}_k$ for $k \leq 9$ at $t = 2, 3, -2, 1/2$.]

There is a useful independent exact test of (16.5). For distinct spectral roots λ_j with degrees d_j , the leading coefficient of P_j can be extracted directly from the sequence F by

$$\text{LC}(P_j) = \frac{\left[\prod_{\ell \neq j} (E - \lambda_\ell)^{d_\ell+1} (E - \lambda_j)^{d_j} F \right] (0)}{\lambda_j^{d_j} d_j! \prod_{\ell \neq j} (\lambda_j - \lambda_\ell)^{d_\ell+1}}. \quad (20.1)$$

The verifier constructs the left-side sequence from Jacobi–Trudi, applies the operator using exact arithmetic, and compares the result with the factorial product (16.5). This checks the signs, the powers of the cluster nodes, and the cross-difference exponents independently of an implementation of the confluent determinant itself.

Finite checks are not a proof of an infinite family. In particular they cannot establish the all-index minimality theorem or historical novelty. Those depend on the mathematical arguments in Sections 14 and 16. No theorem prover was run for this article.

20.2 Dependency and trust boundaries

The main identity depends on the binomial definition, constant terms, finite-dimensional basis changes, block elimination, and dual Jacobi–Trudi. The exact recurrence additionally uses the bialternant, its Taylor confluence, Cauchy–Binet, and finite differences. The numerator theorem uses only those results and an algebraic expansion at infinity. The exceptional-parameter and asymptotic theorems depend on the same explicit leading coefficients. Only Proposition 15.2 imports Schur-module characters and complete reducibility for SL_2 .

Several points are especially important for an independent audit. The Toeplitz pairing is bilinear, not a positive Hermitian inner product; no positivity assumption is used. The odd-order pairing uses the involution $Jf = zf(z^{-1})$, not the ordinary inversion involution. The normalizing constants in the factorization are explicitly fixed at $q = 1$. In the confluence calculation, normalized derivatives remove hidden factorials, and only the highest derivative-order row sets contribute to the maximal degree. Minimality is asserted only after spectral products have been checked for distinctness. Finally, generating functions for $\mathcal{H}_k$ and d_k are kept separate throughout.

20.3 A finite-algebra formalization plan

A Lean implementation can be divided into three layers. First formalize symmetric Laurent coefficient sequences and the two Gram matrices, using finite support; prove the pairing identities and the parameterized Toeplitz factorization. A formal proof can compute basis-change determinants directly or prove their independence from the symbol and use the normalization

specialization as above. All exceptional parameters should be handled by polynomial identity rather than separate division arguments.

The second layer identifies the Toeplitz matrix with the elementary Jacobi–Trudi matrix. Depending on available symmetric-function support, one can initially define the rectangular Schur object by this determinant and separately prove its equality with the complete-homogeneous determinant. This already yields the Hankel identity without a representation-theoretic library.

The third layer formalizes normalized derivative matrices, the factorial Vandermonde determinant (16.13), and degree bounds for Laplace selections. The unique-maximizer argument is finite combinatorics; it is the most important piece to keep inside the trusted proof boundary. The final recurrence follows from explicit shift-operator lemmas. The Python code may generate examples and candidate identities but is not a trusted replacement for any of these steps. No uncompiled Lean skeleton is included as if it were a completed formalization.

21 Further research questions and topics

The following questions are proposed extensions of the results in this article. Except where the original Cigler conjecture is named explicitly, they are not claims about the status of a separately published conjecture.

21.1 Full coefficient structure in Cigler’s Conjecture 17

The spectral polynomials $P_{k,q}(n;t)$ have exact degrees in n , but the present paper does not identify all of their coefficient signs and factorizations after a common denominator is cleared. The numerator polynomials in the source’s Conjecture 17 organize related information in a more refined way. A useful next target is to express those polynomials as flagged tableaux or minors with manifest sign, and to determine exactly which of the printed degree and reciprocity claims survive normalization. The confluent Laplace expansion gives a finite formula from which to start; it is not itself a proof of every Conjecture 17 assertion.

21.2 Complex roots of unity and spectral cancellation

At a complex root of unity, the roots t^q can coincide even when $1, t, t^2$ are distinct. The exact reduced multiplicity is the degree of a sum of canonical $P_{k,q}$ ’s in one residue class, plus one, unless that sum vanishes. Determine a closed formula for these multiplicities as a function of k , the root order, and the residue class. The real points $0, 1, -1$ are settled by Theorem 19.2; simply substituting into the generic product does not solve the complex problem. Schur root-of-unity identities and cyclic-sieving methods may give a structural approach, but cancellation must be checked, not assumed absent.

21.3 Coefficient shape for even shifts

Odd shifts have an SL_2 interpretation that proves parity-wise unimodality. Does the analogous step-two inequality hold for every even shift and every order? If not, what is the first structural obstruction, and what weaker inequalities characterize these tableau weights? The ordinary unimodality assertion is already false, so one should not promote a few unimodal examples into a general conjecture. A satisfactory result would explain the difference between the equal outer multiplicities of odd shifts and the unequal outer multiplicities of even shifts.

21.4 A two-parameter Hankel deformation

The Schur alphabet $((1)^{[a]}, u, (v)^{[b]})$ makes sense for independent u, v . The present moment construction lies on $v = u^2$. Find a natural two-parameter moment family whose normalized shifted Hankel determinants realize the full alphabet, or prove that a specified class of finite-period Jacobi recurrences cannot do so. The parameterized Toeplitz factorization offers a starting point, but the compatibility of all shifts within one moment sequence is a genuine additional condition.

21.5 A uniform crossover as the poles merge

Theorem 19.1 applies for fixed $t < 1$, while at $t = 1$ the answer grows like n^{δ_k} . In the scaling $t = e^{-s/n}$, study

$$n^{-\delta_k} \mathcal{H}_k(n; e^{-s/n})$$

for fixed k and s in bounded complex sets. Determine the limiting function, uniform error bounds, and an all-orders expansion. The finite confluent bialternant should allow a systematic derivation, but the separate spectral coefficients become singular and their cancellations must be organized before taking the limit. This is a natural meeting point of the repository's determinant and asymptotic-analysis programs.

21.6 Broader noncollision criteria and certified algorithms

The middle multiplicity one in $(a, 1, b)$ makes $q = r_2 + 2r_3$ determine the whole allocation. For alphabets with more clusters, classify linear relations among their exponents that still yield a unique allocation, or yield a sum whose leading coefficient can be proved nonzero. Such a criterion would turn Theorem 16.1 into a reusable exact minimal-recurrence algorithm for specialized rectangular Schur families. A formalized implementation should output both the recurrence and a certificate for every pole multiplicity, rather than an annihilator alone.

22 Conclusion

The experimental Schur formula in the inspected ProveIt report admits a finite algebraic proof. Its decisive step is the equality of the original Hankel parity factors with the factors of a particular Toeplitz determinant. The resulting Schur structure supports a sharper conclusion than rationality: all generic recurrence roots and multiplicities are exact, with explicitly nonzero confluent leading coefficients. Negative-width continuation then fixes the numerator degree and reciprocity sign, including a necessary correction to the original printed conjecture. The same calculation gives a sharp convergence equivalent and all reduced denominators at real parameters. The remaining questions concern finer coefficient structure, complex spectral collisions, and uniform pole-merger asymptotics, rather than an unproved bridge in the central determinant identity.

C Small-shift formulas and independent checks

The first three sign-adjusted sequences are

$$\begin{aligned} \mathcal{H}_1(n; t) &= 1, \\ \mathcal{H}_2(n; t) &= 1 + t + \cdots + t^n, \\ \mathcal{H}_3(n; t) &= h_n(1, t, t^2) = \frac{(1 - t^{n+1})(1 - t^{n+2})}{(1 - t)(1 - t^2)}. \end{aligned}$$

The displayed quotient is interpreted by its polynomial continuation at $t = 1, -1$. Their ordinary generating functions are respectively

$$\frac{1}{1-z}, \quad \frac{1}{(1-z)(1-tz)}, \quad \frac{1}{(1-z)(1-tz)(1-t^2z)}.$$

For shift four, (18.9) has numerator degree two, in agreement with $N_4 - 4 = 2$. At $t = -1$ it becomes

$$\sum_{n \geq 0} \mathcal{H}_4(n; -1) z^n = \frac{1+z^2}{(1-z^2)^3},$$

so $\mathcal{H}_4(2m; -1) = (m+1)^2$ and $\mathcal{H}_4(2m+1; -1) = 0$. At $t = 1$, the dimension product gives

$$\mathcal{H}_4(n; 1) = \frac{(n+1)(n+2)^2(n+3)}{12}.$$

These two specializations make visible the difference between a generic recurrence, a pole-merger specialization, and a parity-supported sequence.

D Package contents and reproduction

The archive contains the present `article.tex` and its compiled `article.pdf`, the verifier, exact JSON records, a console log, `README.md`, `STATUS.md`, `sources.md`, and a small `Makefile`. To reproduce the checks and the PDF, run

```
python -m pip install -r requirements.txt
python code/verify.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The recorded run completed all listed checks. Run time is environment-dependent and is not a complexity claim. The package does not include a Lean proof or assert independent referee approval.

[Added 29 September 2026, batch 56: this report ships the verifier, the JSON records (including `data/pdf_checks.json`, a hand-written layout record of the delivered 22-page PDF), the console log, `STATUS.md`, `sources.md`, `requirements.txt` and the `Makefile` under the prefixed names listed in Section 12.1. The archive's `article.tex`, `article.pdf` and `README.md` are not shipped: Part II and the report's `article.pdf` replace them. The commands above are the delivery's and must not be used from this report's root: there `python code/verify.py` runs Part I's verifier, which rewrites Part I's records in `data/`, and `latexmk` rebuilds the whole report in place. The shipped `code/02-schur-recurrences-Makefile` runs exactly these commands (its `clean` target runs `latexmk -c article.tex`), so it must not be run from the report root, and from any other directory its paths do not resolve. Part II's checks are reproduced, from the report root, by

```
uv run --no-project --with sympy==1.14.0 python \
    code/02-schur-recurrences-verify.py --output <scratch>/verification.json
```

with `<scratch>` a directory outside the report.]

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